



Rotor Position and Speed Estimation in a Brushless Direct Current Motor using an Electrical Model and a Disturbance Observer

Josaphat BIFUENI¹, Guy WANLONGO NDIWULU^{1,2}, Angelo KUTI LUSALA²

¹CREI-UK, Faculty of Engineering, Université Kongo (UK), Kongo Central, Democratic Republic of the Congo
josaphatbifueni7@gmail.com/guy.wanlongo@unikin.ac.cd

²Department of Electrical and Computer Engineering, Université de Kinshasa (UNIKIN), Kinshasa, Democratic Republic of the Congo
kutilusala@gmail.com

Corresponding Author: josaphatbifueni7@gmail.com, +243896196013

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Abstract: This paper presents an improved sensorless speed control approach for permanent-magnet brushless DC (BLDC) motors based on back-electromotive force (back-EMF) estimation. The back-EMF is reconstructed using measured phase voltages and currents. While conventional methods rely on the direct differentiation of measured currents—introducing high-frequency noise and phase delays—this approach utilizes a disturbance observer to estimate current derivatives, thereby eliminating noise amplification without increasing computational complexity. Extensive MATLAB/Simulink simulations validate the effectiveness of the method under various operating conditions. Beyond performance comparison, this study includes a detailed sensitivity analysis regarding observer gain tuning (bandwidth ratio and pole placement) and motor parameter mismatches. The results demonstrate that the proposed observer maintains high accuracy, with a position Root Mean Square Error (RMSE) of approximately 8° and a speed Mean Absolute Error (MAE) below 5%, even under parameter variations of up to 120% of nominal resistance and 90% of nominal inductance. These findings define a clear operational envelope for the method and highlight its robustness against thermal drift and magnetic saturation, achieving performance comparable to Hall-effect sensors while establishing the limits for reliable industrial application.

Keywords: Sensorless Approach, Back-EMF Estimation, Disturbance Observer, BLDC Motor, Parameter Robustness

1. INTRODUCTION

A brushless direct current motor (BLDC motor) is a synchronous converter whose rotor consists of permanent magnets, and whose stator windings are supplied with DC voltage through an electronic commutation system [1]. The electronic commutation system plays the same role as the brush commutator system in a conventional DC motor. The commutation times are determined by the rotor position. This position is determined either by sensors such as Hall-effect sensors, encoders, and position resolvers, or by sensorless techniques [2].

There are two main types of brushless DC motors depending on the waveform of the back electromotive force (back-EMF) induced in the stator windings. Back-EMFs may be sinusoidal, as in permanent magnet synchronous motors (PMSM), or trapezoidal, as in permanent magnet brushless DC machines (PMBLDCM). The trapezoidal type is more widely used in many applications due to its simplicity, low cost, high efficiency, and ease of control [3]. The electronic commutation system allows the stator windings to be supplied with variable current. The stator of this motor can be configured as single-phase, two-phase, or three-phase. However, the three-phase motor is the most popular and widely used. For the three-phase motor, the three stator windings can be connected in a star or delta configuration. The star connection is suitable for high torque at low speeds, and the delta connection is suitable for low torque at low speeds [4]. For a star-connected motor, the phase commutation is generally 120° [3], that is, each winding is supplied with a positive voltage for 120° of angular position, and a negative voltage for 120° as well. Over a complete revolution (360° of angular position), each winding is free for 120°. And only two windings are supplied at the same time: one with a positive voltage and the other with a negative voltage, with the third winding floating at that moment.

The BLDC motor has multiple advantages, such as higher efficiency, a higher power-to-weight ratio, a higher torque-to-current ratio, faster dynamic response, reduced maintenance, a better power factor, and higher power output per unit mass and volume, compared to conventional motors [5]. However, the main disadvantages include torque ripple caused by phase switching and the need for rotor position information, which necessitates the use of position sensors. For example, this increases the machine's complexity, causes maintenance problems, and leads to malfunctions in these sensors under certain temperature and pressure conditions. Malfunctioning sensors are unable to accurately provide the rotor position required for proper motor operation [1].

Given the drawbacks associated with position and, in some cases, speed sensors, several sensorless control techniques have been developed for these motors. Based on different sensing principles, sensorless control methods for BLDC motors include back-emf-based, flux-linkage-based, inductance-based, and artificial intelligence-based methods [6, 7].

A commonly adopted solution is described in [8], which relies on the zero-crossing detection of the Back-EMF. This approach is attractive due to its simplicity and low computational burden, making it suitable for low-cost embedded implementations. However, it only provides commutation sequences and does not directly provide rotor speed information, limiting its use to applications where speed regulation is not required. To improve low-speed operation, the method described in [6] introduces an Extended State Observer to overcome the drawback of computing the current derivative in the sensorless estimation chain. Although this strategy enhances the robustness of position estimation at low speeds, it still does not allow rotor speed estimation. A more comprehensive solution is proposed in [9], based on the virtual third-harmonic back-EMF technique, which enables the simultaneous estimation of rotor position and speed. This method achieves accurate rotor angle detection even in the presence of noise and switching disturbances and is particularly suitable for high-speed operation, although its performance decreases at low speeds.

More advanced observer-based and intelligent approaches have also been reported in the literature. The reference [10] combines field-oriented backstepping control with field-oriented input-output feedback linearization, leading to reduced torque ripples and power consumption while improving dynamic response. However, this strategy is more appropriate for PMSM drives than for BLDC motors. Reference [1] presents a sensorless technique based on the Extended Kalman Filter (EKF). This method allows operation over a wide speed range. However, it requires many statistical parameters related to random errors and is highly dependent on motor model accuracy, resulting in a heavy computational load and better suitability for PMSM applications. To overcome these limitations, the approach described in [2] employs a particle filter to estimate both rotor speed and position, offering better robustness than EKF-based methods, at the expense of significantly higher algorithmic complexity. The method reported in [11] uses artificial neural networks, which possess adaptive and self-learning capabilities, a major advantage. Nevertheless, the accuracy of the identification results cannot always be guaranteed. This last approach is applicable to both BLDC and PMSM machines.

Reference [12] presents a simple method that works well across both high- and low-speed ranges. This method reconstructs the back electromotive force (back-EMF) from motor voltage and current measurements using the motor's electrical equation. This Back-EMF contains information about the rotor speed and position. However, this method has a significant drawback: the need to calculate the derivatives of the measured currents. Derivative calculations can be affected by noise in measured signals, leading to calculation errors. Using a filter on the estimated back-EMF signals minimizes the detrimental effects of calculating the current derivative. However, the filter introduces a shift from the actual values of these signals (back-EMF). This shift reduces the accuracy of determining the motor's position and speed during sensorless operation.

This paper proposes an improvement to the method developed in [12], hereafter referred to as the Initial Sensorless Approach, by incorporating a Disturbance Observer to provide the derivative of the measured currents. The new approach, named the Improved Sensorless Approach, estimates the back-EMF using the measured phase voltages and currents of the BLDC motor. In this formulation, the derivative of the phase currents—required by the electrical model for back-EMF estimation—is obtained through a disturbance observer rather than direct numerical differentiation, thereby avoiding noise amplification and phase delay issues.

The theoretical foundations of disturbance observers (DOB), which aim to estimate external disturbances and model uncertainties, have been extensively established in the literature [13]. These methods have since been widely adapted for high-performance motor drives to enhance robustness and precision. In the context of sensorless control, disturbance observers are particularly effective for reconstructing unmeasurable states, such as the back-electromotive force (back-EMF), by treating model mismatches and nonlinearities as lumped disturbances. By integrating these observers into the control loop, it is possible to achieve high-fidelity signal estimation without the noise sensitivity and phase lag typically associated with conventional filtering and numerical differentiation.

The selection of a Standard Disturbance Observer (SDO) for this application is driven by the need for high-precision phase alignment in sensorless control. Unlike Sliding Mode Observers (SMO), which are frequently used in the literature but require additional low-pass filtering to mitigate chattering—thereby introducing undesirable phase delays—the SDO provides a smooth and noise-free estimation of the system states [13]. Furthermore, a hybrid reconstruction strategy is adopted in this paper, rather than direct state estimation. Conventional observer-based methods often exhibit a small inherent phase shift because the error dynamics typically assume a constant back-EMF over a single sampling period. To bypass this limitation, the SDO is employed here as a high-fidelity differentiator to estimate the current derivative. This derivative is then substituted into the motor's fundamental electrical equations to reconstruct the back-EMF. This approach ensures that the reconstructed signal remains perfectly synchronized with the actual rotor position, which is essential for accurate speed.

While robust strategies such as Sliding Mode Observers (SMO) and MRAS are widely used, they often face challenges with chattering-induced delays or complex adaptation laws. Similarly, Extended Kalman Filters (EKF) and recent Machine Learning estimators offer high performance but at the cost of significant computational overhead or 'black-box' opacity. In contrast, the physics-based Disturbance Observer (DOB) approach proposed here, as established by [13], provides a

computationally efficient and transparent alternative, ensuring high-fidelity back-EMF reconstruction without the phase-lag typical of filtered SMO or the complexity of stochastic and AI-based methods.

Beyond the architectural improvements, this study provides an extensive evaluation of the observer's performance. This includes an analysis of the optimal gain tuning strategy—based on bandwidth ratios and non-identical pole placement—as well as a comprehensive sensitivity study regarding stator parameter mismatches. By quantifying the impact of resistance and inductance variations on estimation accuracy, this work defines the operational boundaries of the proposed approach, ensuring its reliability under practical conditions such as thermal drift and magnetic saturation.

The remainder of this paper is organized as follows: Section 2 presents the details of the dynamic model of a permanent-magnet BLDC motor. Section 3 presents the details of the proposed approach for estimating the BLDC motor speed and rotor position, and Section 4 presents the simulation results. Finally, Section 5 concludes the paper.

2. DYNAMIC MODEL OF PERMANENT-MAGNET BLDC MOTOR

The brushless direct current (BLDC) motor is a permanent-magnet synchronous electric machine that lacks the mechanical brush-commutator system found in conventional DC motors. The rotor consists of permanent magnets, while the stator has windings powered by an electronic inverter that performs phase switching. The stator windings can be single-phase or three-phase, as in this paper.

Unlike a conventional DC motor, commutation is performed electronically outside the motor based on the rotor position, resulting in improved reliability, high power density, and superior efficiency [7, 14].

Figure 1 describes a conventional closed-loop control structure of a BLDC motor drive using speed and position sensors. It contains the motor itself, the *BLDC Motor*, the DC voltage source, *DC Supply*, the speed and position sensors, the control system, and the inverter. Mechanical speed and position sensors provide the control system with rotor speed and position signals. The control system converts these signals into switching signals for the inverter switches. The inverter then converts the DC voltage into three-phase voltage to power the three motor windings.

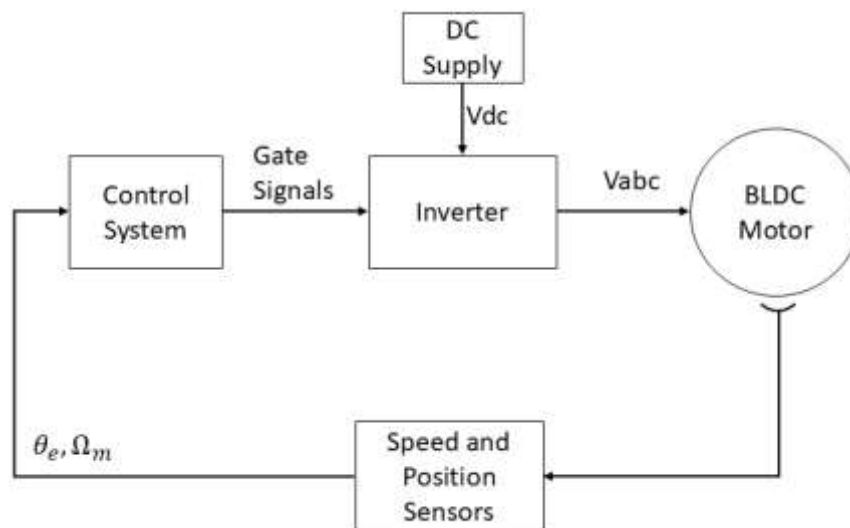


Figure 1: Conventional closed-loop control structure of a BLDC motor drive using speed and position sensors. The DC supply powers the three-phase inverter, which energizes the BLDC motor in response to switching signals generated by the control system. Measured rotor speed and position are fed back to the controller to achieve accurate commutation and speed regulation.

BLDC motors have attracted attention for their longer life, higher efficiency, and lower maintenance requirements compared to conventional brushed motors [14]. They are widely used in applications requiring precise speed and torque control, such as electric vehicles, drones, robotics, and industrial systems, providing high torque with minimal energy consumption and enabling sustainable operation [5].

In BLDC motors, each back electromotive force (back-EMF) exhibits a trapezoidal wave, and the mutual inductance between the stator and rotor is non-sinusoidal [15]. Since the three back electromotive forces of a permanent-magnet BLDC motor are trapezoidal, it is more appropriate to develop the model in phase variables rather than in the d-q synchronous reference frame. The developed model is based on the following assumptions:

- Currents induced in the rotor by the stator harmonic fields are neglected;
- Magnetic losses and iron losses are neglected;
- Resistances of all stator windings are equal, and the self- and mutual inductances of these windings are constant;
- Magnetic circuit is not saturated.

Figure 2 gives an electrical diagram of the BLDC motor. Each phase is represented by a back-EMF voltage source e_i in series with the phase resistance R_i and inductance L_i and is supplied by the corresponding inverter phase voltage v_i . Mutual inductances M are included to account for the electromagnetic coupling between the motor windings. i denotes phases a, b and c.

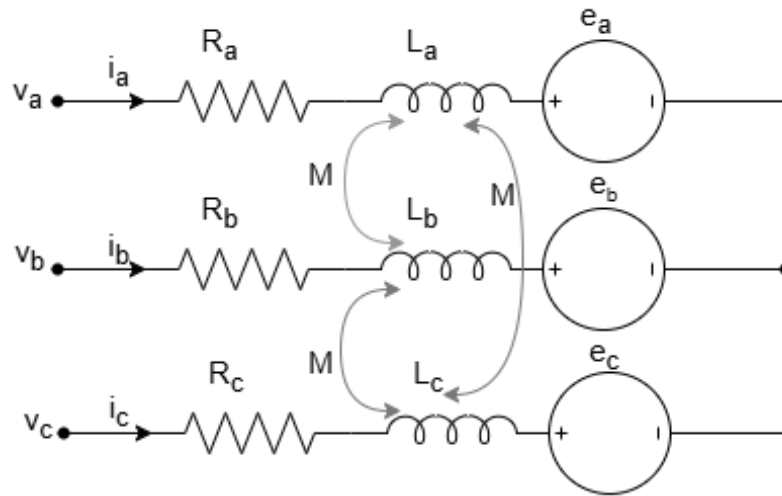


Figure 2: Equivalent electrical model of a three-phase BLDC motor with star-connected stator windings.

A dynamic system is generally modeled using a set of nonlinear differential and algebraic equations. In its most general continuous form, the dynamic model is written using the following state representation [16]:

$$\dot{x}(t) = f(x, u, t) \quad (1)$$

$$y(t) = g(x, u, t) \quad (2)$$

Equation (1) is the state equation and Equation (2) is the output equation. The nonlinear functions f and g , respectively, characterize the internal dynamics and the output relation of the system. $x(t)$ is the state vector describing the internal dynamics of the system. u is the vector of command inputs. $y(t)$ is the vector of measured outputs.

Let's consider that equations (1) and (2) are linearized around the operating point. This involves the equations given by (3) and (4). Where $A(t)$ represents the dynamic matrix of the considered dynamic system. $B(t)$ is the command matrix. $C(t)$ is the output matrix. $D(t)$ denotes the direct transmission matrix:

$$\dot{x}(t) = A(t)x(t) + B(t)u(t) \quad (3)$$

$$y(t) = C(t)x(t) + D(t)u(t) \quad (4)$$

If vector functions f and g do not involve time t explicitly then the system is called a time-invariant system. In this case, Equations (3) and (4) can be simplified to:

$$\dot{x}(t) = Ax(t) + Bu(t) \quad (5)$$

$$y(t) = Cx(t) + Du(t) \quad (6)$$

Dynamic modeling of the BLDC motor aims to mathematically describe the time evolution of the system's electrical and mechanical quantities, such as phase currents, speed, and rotor position. This modeling is based on the electrical equations of the stator windings, incorporating resistance, inductance, and back electromotive force (back-EMF), as well as the mechanical equation relating electromagnetic torque to rotor dynamics. The dynamic model is a fundamental tool for designing control laws, analyzing transient behavior, and developing sensorless techniques for estimating speed and position without mechanical sensors.

The BLDC motor model consists of two parts: the electrical and mechanical models, presented in the following sections.

2.1 Electrical Model

The electrical dynamic model, as described in Figure 2, can be modelled by the following equations:

$$\begin{cases} v_a = Ri_a + L \frac{di_a}{dt} + M \frac{di_b}{dt} + M \frac{di_c}{dt} + e_a \\ v_b = Ri_b + L \frac{di_b}{dt} + M \frac{di_a}{dt} + M \frac{di_c}{dt} + e_b, \\ v_c = Ri_c + L \frac{di_c}{dt} + M \frac{di_a}{dt} + M \frac{di_b}{dt} + e_c \end{cases} \quad (7)$$

where v_a , v_b and v_c are the phase voltages. i_a , i_b and i_c are respectively stator currents in phases a, b and c. e_a , e_b and e_c represent respectively back-electromotive forces induced in the windings a, b and c.

Since the stator windings are star-coupled, the stator currents are related by the relationship:

$$i_a + i_b + i_c = 0 \quad (8)$$

Therefore, Equation (7) can be rewritten as:

$$\begin{cases} v_a = Ri_a + (L - M) \frac{di_a}{dt} + e_a \\ v_b = Ri_b + (L - M) \frac{di_b}{dt} + e_b, \\ v_c = Ri_c + (L - M) \frac{di_c}{dt} + e_c \end{cases} \quad (9)$$

The back-electromotive forces (back-EMF) induced in the windings can be written in the form:

$$\begin{cases} e_a = \frac{p\omega_m 3\psi_0}{\pi} f_a(\theta_r) = \lambda_p \omega_e f_a(\theta_e) = E_p f_a(\theta_e) \\ e_b = \frac{p\omega_m 3\psi_0}{\pi} f_b(\theta_r) = \lambda_p \omega_e f_b(\theta_e) = E_p f_b(\theta_e), \\ e_c = \frac{p\omega_m 3\psi_0}{\pi} f_c(\theta_r) = \lambda_p \omega_e f_c(\theta_e) = E_p f_c(\theta_e) \end{cases} \quad (10)$$

where θ_e represents the electrical angle of the rotor. $f_a(\theta_e)$, $f_b(\theta_e)$ and $f_c(\theta_e)$ are the functions that respectively represent the shape of the back-EMF e_a , e_b and e_c , but with a unit amplitude. λ_p and E_p denote the magnitude of the flux linkage and the magnitude of the back-EMF, respectively. They can be expressed by:

$$\lambda_p = \frac{3\psi_0}{\pi} \quad (11)$$

$$E_p = \lambda_p \omega_e = K_e \omega_m \quad (12)$$

where ω_m is the mechanical speed of rotation of the rotor in radians per second. $\omega_e = p\omega_m$ is the electrical speed of the motor. $K_e = p\lambda_p$, it represents the back-EMF constant. p denotes the number of pole pairs. θ_e denotes the electrical angle of the rotor position in radians. The instantaneous electrical angle θ_e is related to the instantaneous rotor's mechanical position θ_m by the number of pole pairs of the rotor magnets ($\theta_e = p\theta_m$).

Equation (12) shows that the back-emf amplitude is proportional to the rotor speed. It is this proportionality relationship that the sensorless method presented in the following sections exploits to extract the rotor speed from the back-EMF amplitude.

Figure 3 shows the trapezoidal waveforms of the back-EMFs relative to the electrical position of the rotor. The electrical angle on the horizontal axis highlights the 120° phase-shifted trapezoidal nature of the back-EMF signals. These waveforms directly define the commutation logic and indicate the rotor position intervals at which each stator winding must be energized. $f_a(\theta_e)$, $f_b(\theta_e)$ and $f_c(\theta_e)$ have the same appearance, but with unit amplitude. e_a , e_b and e_c represent the back-EMF of windings a, b and c respectively.

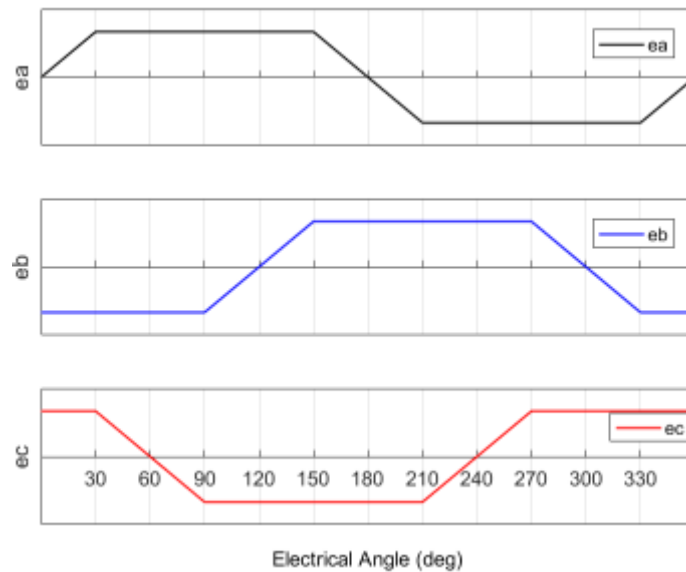


Figure 3: Trapezoidal back-EMF waveforms of the three-phase BLDC motor as a function of the rotor electrical position.

The waveform of the back-EMF is very important in deciding the motor control. For winding a for example, the back-emf e_a is positive and constant from 30° to 150° . During this period, the inverter must supply a positive voltage to circulate a positive current through this winding to produce positive electromagnetic power. e_a is negative and constant from 210° to 330° , and the inverter must supply a negative voltage to produce positive electromagnetic power. It is this electromagnetic power that is linked to the motor's electromagnetic torque, responsible for the rotor's rotational movement.

The electromagnetic power is expressed by Equation (13):

$$P_{em} = (e_a i_a + e_b i_b + e_c i_c) \quad (13)$$

2.2 Mechanical Model

The rotor's mechanical rotational speed ω_m is given by:

$$\omega_m = \frac{d\theta_m}{dt} \quad (14)$$

The electromagnetic torque T_e , for a given rotation speed, is given by:

$$T_e = \frac{P_{em}}{\omega_m} = (e_a i_a + e_b i_b + e_c i_c) \frac{1}{\omega_m} \quad (15)$$

Equation (15) can be rewritten as:

$$T_e = \lambda_p [f_a(\theta_e) i_a + f_b(\theta_e) i_b + f_c(\theta_e) i_c] \quad (16)$$

The swing equation of motion of a dynamic system with J as the moment of inertia, B as the coefficient of friction, and T_l the resistive torque of the load, is:

$$J \frac{d\omega_m}{dt} + B\omega_m = (T_e - T_l) \quad (17)$$

2.3 Complete Dynamic model

Dynamic modelling of the DC motor can consist of the sub-electrical and sub-mechanical dynamic models. The electrical and mechanical equations are combined to form the dynamic model given in equation (3). State model has the following form:

$$\dot{x} = Ax + Bu \quad (18)$$

Equation (18) represents state model of motor. The system's state variables x , the matrix of system's dynamic and matrix B are given by:

$$x = \begin{bmatrix} i_a \\ i_b \\ i_c \\ \omega_m \\ \theta_e \end{bmatrix}, A = \begin{bmatrix} -\frac{R}{L_m} & 0 & 0 & -\frac{\lambda_p}{L_m} f_a(\theta_e) & 0 \\ 0 & -\frac{R}{L_m} & 0 & -\frac{\lambda_p}{L_m} f_b(\theta_e) & 0 \\ 0 & 0 & -\frac{R}{L_m} & -\frac{\lambda_p}{L_m} f_c(\theta_e) & 0 \\ \frac{\lambda_p}{J} f_a(\theta_e) & \frac{\lambda_p}{J} f_b(\theta_e) & \frac{\lambda_p}{J} f_c(\theta_e) & -\frac{B}{J} & 0 \\ 0 & 0 & 0 & p & 0 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1/L_m & 0 & 0 & 0 \\ 0 & 1/L_m & 0 & 0 \\ 0 & 0 & 1/L_m & 0 \\ 0 & 0 & 0 & -1/J \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

This system dynamic model describes the complete dynamic model of a three-phase BLDC motor with stator windings connected in a star configuration. This dynamic model is described in Figure 4. The input signals are the three-phase voltage v_{abc} and the load torque. The difference between the three-phase voltage signals v_{abc} and the three-phase back-EMF signals e_{abc} is calculated using a transfer function representing dynamic model of the electrical part to provide current signals i_{abc} . The current i_{abc} , the linkage flux, and the rotor position are the inputs for calculating the motor's electromagnetic torque. The difference between the motor torque and the load torque is calculated using a transfer function that represents the mechanical part's dynamic model to determine the rotor's mechanical speed. The rotor's mechanical position is found by integrating the rotor speed. The rotor's electrical position is the product of the mechanical position and the number of pole pairs. The electrical position, the rotor's mechanical speed, and the linkage flux are inputs to the back-EMF calculation block e_{abc} .

3. PROPOSED APPROACH TO ESTIMATE SPEED AND ROTOR POSITION

In the classical control method for a BLDC motor, the rotor speed and position are detected by speed and position sensors, respectively. This is called direct speed-detection and position-detection methods. Under certain operating conditions, position and speed sensors can experience accuracy issues due to noise and measurement errors. These problems are caused by extreme temperatures, dust, wear and tear, aging, and humidity as described in [14]. These problems require maintenance, and they increase the complexity of the electric motor. To improve these problems, sensorless methods are developed as described in [7]. These sensorless methods use voltage and current signals to detect

rotor position and/or rotor speed. These sensorless methods use voltage and current signals to detect rotor position and/or rotor speed. The voltage and current signals are used in these methods because these signals are easy to acquire, even under difficult motor operating conditions. These sensorless approaches are called indirect rotor-position-detection methods and indirect rotor-speed-detection methods.

This paper focuses on indirect rotor-position and rotor-speed detection methods based on motor back-EMF estimation. The motor back-EMF signal contains speed and position information. The detailed development of both methods is given in the following sections.

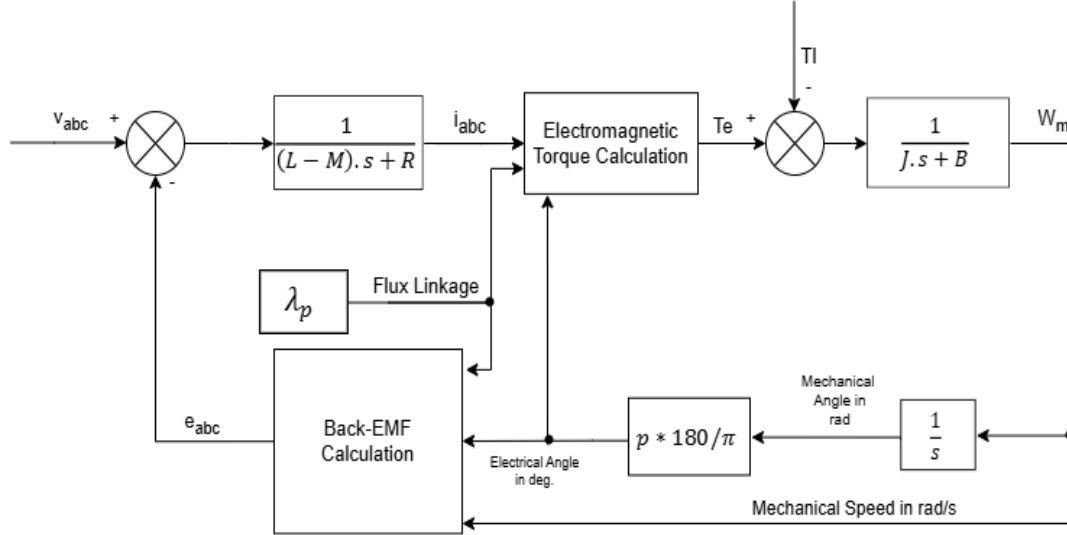


Figure 4: Complete dynamic model of the BLDC motor describing the electrical and mechanical subsystems. The model takes the three-phase stator voltage v_{abc} and the load torque T_l as inputs and computes the phase currents, electromagnetic torque, rotor speed, and rotor position through interconnected electrical and mechanical blocks. The electrical position and the resulting three-phase back-EMF e_{abc} are continuously updated based on the rotor motion and flux linkage.

3.1 Approach Based on Direct Reconstruction of the Back-EMF

This approach is proposed in [12]. It allows the motor's back-EMF to be calculated from the current and voltage signals of each phase using the BLDC motor's dynamic model voltage equation. Once the back-EMF of 3 windings has been calculated, their trapezoidal shape is used to extract the position of the rotor, and their amplitude will be used to extract the speed of the rotor. In this article, this approach will be referred to as the **Initial Sensorless Approach**.

The dynamic model of the motor can be described by the following equations, where the motor parameters are the resistances and the self- and mutual inductances of the windings.

$$v_{abc} = e_{abc} + \left(Ri_{abc} + L_m \frac{di_{abc}}{dt} \right) \quad (19)$$

where v_{abc} , e_{abc} , R , L and i_{abc} are matrices which are given by:

$$v_{abc} = \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix}, e_{abc} = \begin{bmatrix} e_a \\ e_b \\ e_c \end{bmatrix}, R = \begin{bmatrix} R_s & 0 & 0 \\ 0 & R_s & 0 \\ 0 & 0 & R_s \end{bmatrix}, L = \begin{bmatrix} L_m & 0 & 0 \\ 0 & L_m & 0 \\ 0 & 0 & L_m \end{bmatrix} \text{ and } i_{abc} = \begin{bmatrix} i_a \\ i_b \\ i_c \end{bmatrix}$$

From Equation (19), the back-electromotive force can be expressed as:

$$e_{abc} = v_{abc} - \left(Ri_{abc} + L_m \frac{di_{abc}}{dt} \right) \quad (20)$$

The electrical model of the motor showed that the back-EMF has a periodic trapezoidal shape, with amplitude directly proportional to speed via the constant K_e . By dividing the back-EMF e_{abc} by the constant K_e , a function y_a can be defined having the same trapezoidal shape as e_{abc} , but with an amplitude equal to the rotor speed. In three-phase notation, the y_{abc} function is given by:

$$y_{abc} = \frac{e_{abc}}{K_e} = \begin{bmatrix} y_a \\ y_b \\ y_c \end{bmatrix} \quad (21)$$

Equation (21) shows that the function y_{abc} has the same form as the back-EMF e_{abc} because K_e is constant. Its magnitude equals the rotor's electrical speed, and its trapezoidal shape provides information about the rotor's position.

Figure 5 shows the waveform of the functions y_a , y_b and y_c . The functions y_a , y_b and y_c preserve the trapezoidal shape of the back-EMF, while their amplitude is normalized by the back-EMF constant, making them directly proportional to the

electrical rotor speed ω_r . This property shows that the motor speed can be obtained by extracting the amplitude of y_{abc} using the proposed sensorless method based on measured phase voltages v_{abc} and currents i_{abc} .

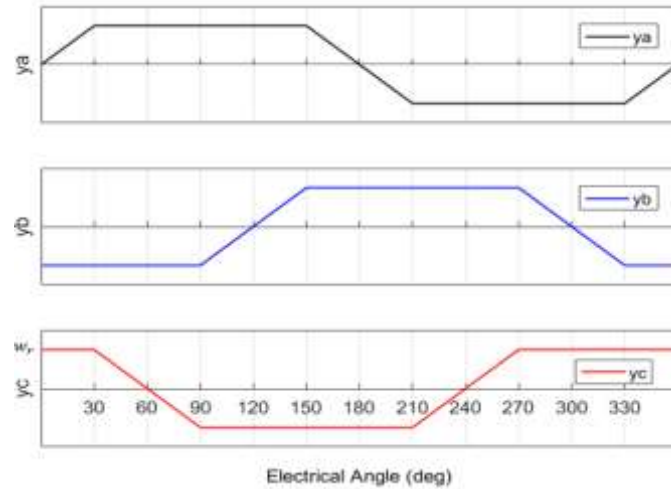


Figure 5: Three-phase trapezoidal function y_{abc} (y_a , y_b and y_c) obtained from the estimated back-EMF e_{abc} .

The electrical speed of the motor ω_r , being the magnitude of the function y_{abc} , can be calculated by:

$$\omega_r = \max(y_a, y_b, y_c) \quad (22)$$

If the speed is known, the rotor position can be estimated using the following Table 1 for $\omega_r > 0$ and Table 2 for $\omega_r < 0$.

Table 1 maps the rotor angle to the trapezoidal shape of the functions y_a , y_b and y_c , as shown in Figure 2. The rotor position is determined from the trapezoidal shape of these functions, whose amplitudes are equal to the electrical motor speed ω_r , leading to normalized expressions of the form y_a/ω_r , y_b/ω_r , and y_c/ω_r . The position estimation is defined over 12 successive intervals of $\pi/6$ radians, each associated with a specific analytical expression for positive speed. For example, in the interval $[0, \pi/6]$, y_b is constant and negative. While y_c is constant and positive. Only y_a increases. The rotor position is thus proportional to y_a and inversely proportional to the speed. Indeed, since the magnitude of y_a has the same unit as the rotor speed. The ratio y_a/ω_r yields a dimensionless number, which is converted to radians by multiplying by π .

Table 2 does the same thing, but for a negative speed. The rotor position is determined from the trapezoidal shape of these functions, whose amplitudes are equal to the electrical motor speed ω_r , leading to normalized expressions in the form y_a/ω_r , y_b/ω_r , and y_c/ω_r . The position estimation is defined over twelve successive intervals of $\pi/6$ radians, each associated with a specific analytical expression for negative speed.

Table 1: Relationship between the rotor electrical position and the functions y_a , y_b , and y_c of positive speed

No.	Condition	Estimated electrical position of the rotor θ_e (rad)
1	$y_b < y_a < y_c$ & $y_a > 0$	$\theta_e = \alpha\pi y_a/\omega_r$
2	$y_b < y_c < y_a$ & $y_c > 0$	$\theta_e = \pi/3 - \alpha\pi y_c/\omega_r$
3	$y_b < y_c < y_a$ & $y_c < 0$	$\theta_e = \pi/3 - \alpha\pi y_c/\omega_r$
4	$y_c < y_b < y_a$ & $y_b < 0$	$\theta_e = 2\pi/3 + \alpha\pi y_b/\omega_r$
5	$y_c < y_b < y_a$ & $y_b > 0$	$\theta_e = 2\pi/3 + \alpha\pi y_b/\omega_r$
6	$y_c < y_a < y_b$ & $y_a > 0$	$\theta_e = \pi - \alpha\pi y_a/\omega_r$
7	$y_c < y_a < y_b$ & $y_a < 0$	$\theta_e = \pi - \alpha\pi y_a/\omega_r$
8	$y_a < y_c < y_b$ & $y_c < 0$	$\theta_e = 4\pi/3 - \alpha\pi y_c/\omega_r$
9	$y_a < y_c < y_b$ & $y_c > 0$	$\theta_e = 4\pi/3 - \alpha\pi y_c/\omega_r$
10	$y_a < y_b < y_c$ & $y_b > 0$	$\theta_e = 5\pi/3 - \alpha\pi y_b/\omega_r$
11	$y_a < y_b < y_c$ & $y_b < 0$	$\theta_e = 5\pi/3 - \alpha\pi y_b/\omega_r$
12	$y_b < y_a < y_c$ & $y_a < 0$	$\theta_e = 2\pi + \alpha\pi y_a/\omega_r$

And, for $\omega_r < 0$, the rotor position can be estimated according the following Table 2:

Table 2: Relationship between the rotor electrical position and the functions y_a , y_b , and y_c for negative rotor speed.

No.	Condition	Estimated electrical position of the rotor θ_e (rad)
1	$y_c < y_a < y_b$ & $y_a < 0$	$\theta_e = \alpha\pi y_a / \omega_r$
2	$y_a < y_c < y_b$ & $y_c < 0$	$\theta_e = -\pi/3 - \alpha\pi y_c / \omega_r$
3	$y_a < y_c < y_b$ & $y_c > 0$	$\theta_e = -\pi/3 - \alpha\pi y_c / \omega_r$
4	$y_a < y_b < y_c$ & $y_b > 0$	$\theta_e = -2\pi/3 + \alpha\pi y_b / \omega_r$
5	$y_a < y_b < y_c$ & $y_b < 0$	$\theta_e = -2\pi/3 + \alpha\pi y_b / \omega_r$
6	$y_b < y_a < y_c$ & $y_a < 0$	$\theta_e = -\pi - \alpha\pi y_a / \omega_r$
7	$y_b < y_a < y_c$ & $y_a > 0$	$\theta_e = -\pi - \alpha\pi y_a / \omega_r$
8	$y_b < y_c < y_a$ & $y_c > 0$	$\theta_e = -4\pi/3 - \alpha\pi y_c / \omega_r$
9	$y_b < y_c < y_a$ & $y_c < 0$	$\theta_e = -4\pi/3 - \alpha\pi y_c / \omega_r$
10	$y_c < y_b < y_a$ & $y_b < 0$	$\theta_e = -5\pi/3 - \alpha\pi y_b / \omega_r$
11	$y_c < y_b < y_a$ & $y_b > 0$	$\theta_e = -5\pi/3 - \alpha\pi y_b / \omega_r$
12	$y_b < y_a < y_c$ & $y_a > 0$	$\theta_e = -2\pi + \alpha\pi y_a / \omega_r$

The coefficient α is used to adjust the trapezoidal profile of the estimated back-EMF via the function y_{abc} , specifically to compensate for the phase shift that may occur between the actual and estimated rotor positions. While its default value is set to 1, the impact of this parameter on the sensorless method's performance was evaluated by varying it from 0.5 to 2. The results indicate that to maintain position estimation accuracy, α must remain within a specific operational range: it must be greater than or equal to 1 and cannot exceed a critical threshold of 1.5, beyond which the synchronization between the estimated and real positions significantly deteriorates.

Figure 6 depicts the model of the Initial Sensorless Approach based on direct reconstruction of the Back-EMF. The sensorless block directly differentiates the measured phase currents and combines the resulting derivatives with measured voltages and currents to estimate the back-EMF. The extracted rotor speed and position replace mechanical sensors and are fed back to the control. The proposed approach is associated with the BLDC motor, which receives three-phase voltage signals v_{abc} , it generates three-phase current signals i_{abc} . Voltage signals are provided by a voltage-source inverter, which is controlled by gate signals from the control system for the DC motor's rotor position θ_e and speed Ω_m . The back-EMF calculation block receives voltage, current, and derivative current signals to compute the back-electromotive force $\hat{e}_{abc}$. The three-phase back-EMF is used to calculate rotor position and speed, which are used by the control system. The derivative current signals $\frac{di_{abc}}{dt}$ are provided by a digital device which receives the measured currents as input.

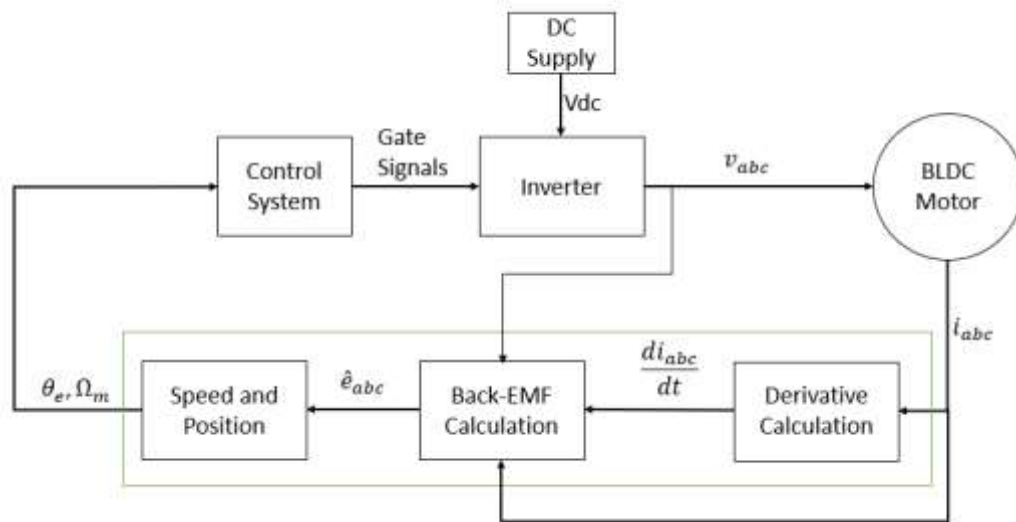


Figure 6: Sensorless BLDC motor control scheme based on the Initial Sensorless Approach

This sensorless approach relies on measuring stator voltages and currents and computing their derivatives using conventional digital differentiation techniques. Obtaining this current derivative is a significant source of performance degradation. These methods strongly amplify the high-frequency components present in the current signals, which result from measurement noise, electromagnetic interference, and inverter switching effects [6]. This amplification leads to rapid fluctuations and large-amplitude peaks in the estimated current derivative, which propagate directly through the sensorless

estimation chain, as described in [6]. Consequently, the quality of the signals used to reconstruct the back-EMF is severely degraded, compromising the accuracy of the rotor position and speed estimation.

To mitigate these disturbances, the use of low-pass filters applied to the estimated back-EMF is generally considered [12]. However, while these filters significantly reduce peaks and high-frequency noise, they inevitably introduce a phase lag between the motor's actual back-EMF and its estimate. This phase shift introduces a systematic delay in the estimated rotor position, leading to suboptimal stator current commutation. Consequently, the stator field is no longer properly aligned with the rotor flux, resulting in a decrease in effective electromagnetic torque and an overall degradation of dynamic performance. Thus, for the same supply voltage and load, a motor driven from the estimated position exhibits a lower rotational speed than one obtained with direct measurement or an ideal estimate, as illustrated in the simulation results in Figure 13. This demonstrates the negative impact of calculating the current derivative on the efficiency and accuracy of this sensorless approach.

To overcome the limitations inherent in directly calculating the derivative of the measured current in this sensorless approach, a Disturbance Observer was integrated into the estimation chain, as described in the following section.

3.2 Proposed Method to Improve the Estimation of Rotor Speed and Rotor Position

The disturbance observer method uses only the measured stator voltages and currents to robustly reconstruct the motor's electrical dynamics, providing an estimate of the current derivative. Unlike the conventional derivative operator, the observer serves as an intrinsic dynamic filtering mechanism, effectively rejecting measurement noise, PWM switching-induced disturbances, and parametric uncertainties. The observer estimates the current derivative, which is then fed back into the Initial Sensorless Approach, thereby improving the back-EMF estimation strategy. In this article, this approach will be referred to as the **Improved Sensorless Approach**. The design of this disturbance observer method is proposed in [17].

The voltage equation can be rewritten as follows:

$$\begin{cases} \frac{di_a}{dt} = a_{11}i_a + a_{12}e_b + b_1v_a \\ \frac{di_b}{dt} = a_{11}i_b + a_{12}e_b + b_1v_b \\ \frac{di_c}{dt} = a_{11}i_c + a_{12}e_c + b_1v_c \end{cases} \quad (23)$$

where: $a_{11} = -\frac{R}{L-M} = -\frac{R}{L_m}$, $a_{12} = -\frac{1}{L-M} = -\frac{1}{L_m}$ and $b_1 = \frac{1}{L-M} = \frac{1}{L_m}$

By the simplification hypothesis mentioned above, the back-EMF is assumed to be a constant disturbance, $\dot{e}_a = 0$, $\dot{e}_b = 0$, and $\dot{e}_c = 0$.

Thus, the state variable model of the BLDC motor for each phase is:

$$\begin{bmatrix} \dot{i}_x \\ \dot{e}_x \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} i_x \\ e_x \end{bmatrix} + \begin{bmatrix} b_1 \\ 0 \end{bmatrix} v_x \quad (24)$$

$$y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} i_x \\ e_x \end{bmatrix} \quad (25)$$

in which, phase voltage v_x is the input variable, current i_x is the output variable, and back-EMF e_x is imposed on the system as a disturbance.

Thus, the full-order state observer can be designed as (for each phase):

$$\begin{aligned} \frac{d}{dt} \begin{bmatrix} \hat{i}_x \\ \hat{e}_x \end{bmatrix} &= \begin{bmatrix} a_{11} & a_{12} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \hat{i}_x \\ \hat{e}_x \end{bmatrix} + \begin{bmatrix} b_1 \\ 0 \end{bmatrix} v_x \\ &\quad + \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} (y - \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \hat{i}_x \\ \hat{e}_x \end{bmatrix}) \end{aligned} \quad (26)$$

$$\begin{cases} \hat{i}_x(0) = 0 \\ \hat{e}_x(0) = 0 \end{cases} \quad (27)$$

where g_1 and g_2 are the feedback gain parameters of the full-state observer for phase 'x'.

This equation can be arranged as follows:

$$\frac{d}{dt} \begin{bmatrix} \hat{i}_x \\ \hat{e}_x \end{bmatrix} = \begin{bmatrix} a_{11} - g_1 & a_{12} \\ -g_2 & 0 \end{bmatrix} \begin{bmatrix} \hat{i}_x \\ \hat{e}_x \end{bmatrix} + \begin{bmatrix} b_1 \\ 0 \end{bmatrix} v_x + \begin{bmatrix} g_1 \\ g_2 \end{bmatrix} y \quad (28)$$

1) Error dynamics:

Define the estimation errors:

$$\begin{aligned}\tilde{i}_x &= i_x - \hat{i}_x \\ \tilde{e}_x &= e_x - \hat{e}_x\end{aligned}\quad (29)$$

Subtracting the observer equations from the real system equations yields the error dynamics:

$$\begin{aligned}\dot{\tilde{i}}_x &= (a_{11} - g_1)\tilde{i}_x + a_{12}\tilde{e}_x \\ \dot{\tilde{e}}_x &= -g_2\tilde{i}_x + \dot{e}_x\end{aligned}\quad (30)$$

Since the back-EMF e_x is assumed to be a constant disturbance, $\dot{e}_x = 0$.

In matrix form, error dynamics can be expressed as:

$$\frac{d}{dt} \begin{bmatrix} \tilde{i}_x \\ \tilde{e}_x \end{bmatrix} = \begin{bmatrix} a_{11} - g_1 & a_{12} \\ -g_2 & 0 \end{bmatrix} \begin{bmatrix} \tilde{i}_x \\ \tilde{e}_x \end{bmatrix}\quad (31)$$

The disturbance observer approach provides a current derivative, enabling the sensorless method to avoid the disadvantages of current derivative calculation. The observer's stability analysis will determine its gains. The stability of a linear system is entirely determined by the position of its poles in the complex plane. It must be ensured that the poles are negative real numbers for the system to be stable and for the estimation error to tend to 0 rapidly.

The characteristic polynomial of the error system matrix A_e is given (for a phase):

$$p(s) = \det(sI - A_e) = s^2 + (g_1 - a_{11})s + a_{12}g_2\quad (32)$$

2) Stability conditions:

For a second-order system, asymptotic stability is guaranteed if the polynomial is Hurwitz [16]. Therefore, the following conditions must be satisfied:

$$g_1 - a_{11} > 0\quad (33)$$

$$a_{12}g_2 > 0\quad (34)$$

Since $a_{11} = -\frac{R}{L_m} < 0$ and $a_{12} = -\frac{1}{L_m} < 0$, the stability conditions become:

$$g_1 > -\frac{R}{L_m}\quad (35)$$

$$g_2 < 0\quad (36)$$

Hence, the Disturbance Observer is stable for any gains g_1 and g_2 satisfying these inequalities.

3) Gain selection using pole placement:

To ensure fast convergence of the estimated current derivative, the observer poles may be placed at $-\omega_0$ with a double pole assumption [18]:

$$(s + \omega_0)^2 = s^2 + 2\omega_0 s + \omega_0^2\quad (37)$$

By identification with the observer characteristic polynomial, the gains are obtained as:

$$g_1 = a_{11} + 2\omega_0 = -\frac{R}{L_m} + 2\omega_0\quad (38)$$

$$g_2 = \frac{\omega_0^2}{a_{12}} = -\omega_0^2 L_m\quad (39)$$

Where ω_0 represents the observer bandwidth.

When the observer gains are selected using identical poles $(s + \omega_0)^2$, the resulting dynamics may not always achieve the desired estimation performance. In practice, this choice can lead to an unfavorable trade-off between convergence speed and noise sensitivity, especially when the estimation of the extended state (back-EMF) requires a faster correction action than the current state. Therefore, a more flexible approach consists in assigning two distinct observer poles.

The desired characteristic polynomial is defined as:

$$(s + p_1)(s + p_2) = s^2 + (p_1 + p_2)s + p_1 p_2\quad (40)$$

By identification, the observer gains are obtained as:

$$g_1 = a_{11} + (p_1 + p_2) = -\frac{R}{L_m} + (p_1 + p_2) \quad (41)$$

$$g_2 = \frac{p_1 p_2}{a_{12}} = -L_m p_1 p_2 \quad (42)$$

4) Practical range of observer bandwidth:

The choice of ω_0 is a compromise between convergence speed and measurement noise sensitivity. A small ω_0 leads to slow estimation dynamics, while a large ω_0 increases sensitivity to current measurement noise and inverter switching ripple. In practice, ω_0 can be selected in the range:

$$\omega_0 = (5 \sim 20) \omega_e \quad (43)$$

Where ω_e is the electrical angular speed of the motor. This range ensures adequate robustness and fast transient response.

To assign two distinct observer poles, one may define for instance:

$$p_1 = \omega_0 \quad (44)$$

$$p_2 = k \omega_0$$

where k is a positive real constant ($k > 0$). In this case, the observer gains become :

$$g_1 = -\frac{R}{L_m} + (1 + k) \omega_0 \quad (45)$$

$$g_2 = \frac{p_1 p_2}{a_{12}} = -L_m k \omega_0^2 \quad (46)$$

This formulation provides a family of observer gains parameterized by k , allowing a flexible adjustment of observer dynamics while maintaining stability.

According to recent literature on high-performance BLDC drives [19], a Root Mean Square Error (RMSE) for position below 8° electrical is required to ensure optimal torque production and minimize ripple. Regarding speed estimation, while a Mean Absolute Error (MAE) within 1% of the nominal range is the benchmark for high-precision applications, a deviation of up to 5% is generally considered acceptable in industrial contexts, particularly when the system is subjected to significant parameter variations or extreme load disturbances.

Figure 7 depicts the disturbance observer model (for a phase). Using measured phase voltages and currents, the observer estimates current derivatives without directly differentiating noisy signals. The estimated derivatives serve as inputs to the back-EMF estimation process. In this figure, the voltage v_x and current i_x of one phase of the motor serve as inputs to the observer. This observer implements equation (28) and provides the current derivative $\frac{d}{dt} \hat{i}_x$ as its output variable.

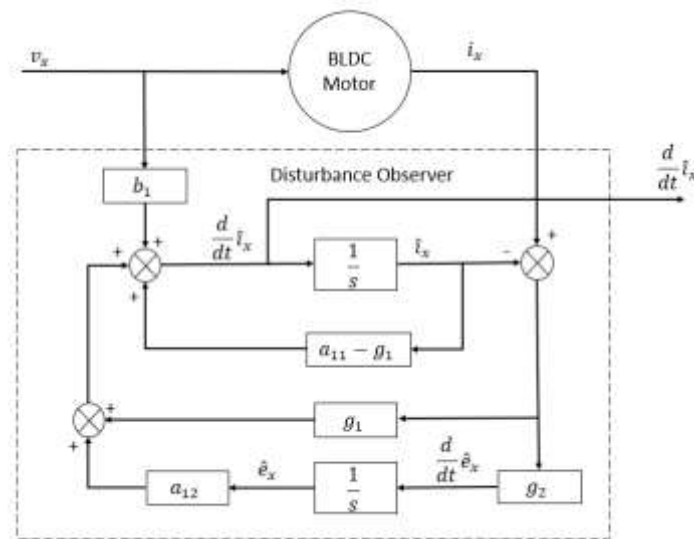


Figure 7: Disturbance observer structure for phase current derivative estimation in a sensorless BLDC motor drive.

Figure 8 depicts a complete sensorless BLDC motor control scheme based on the Improved Sensorless Approach. Unlike the initial approach, the phase current derivatives are estimated using a disturbance observer fed by measured phase

voltages and currents, rather than direct numerical differentiation. The estimated current derivatives, together with the measured voltages and currents, are injected into the back-EMF calculation block, while the remaining control structure is identical to that of Figure 6. The proposed method is associated with the BLDC motor, which receives three-phase voltage signals v_{abc} , and it generates three-phase current signals i_{abc} . Voltage signals are provided by a voltage-source inverter, which is controlled by gate signals from the control system for the BLDC motor's rotor position θ_e and speed ω_m . The back-EMF calculation block receives voltage, current, and derivative current signals to compute the back-electromotive force $\hat{e}_{abc}$. The three-phase back-EMF is used to calculate rotor position and speed, which are used by the control system. The derivative current signals $\frac{di_{abc}}{dt}$ are provided by a Disturbance Observer that receives the measured three-phase currents and three-phase voltages as inputs.

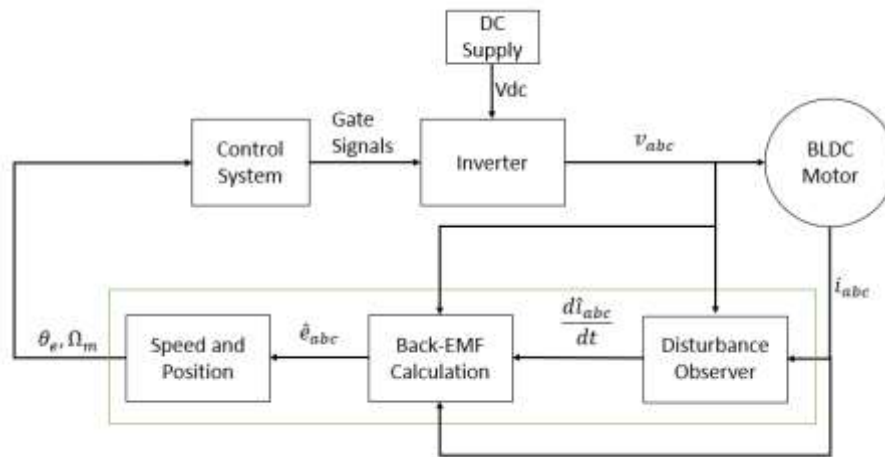


Figure 8: Complete sensorless BLDC motor control scheme based on the Improved Sensorless Approach.

This figure, therefore, shows that the observer is used solely to compute the current derivative from the voltage and current measurements. The rest of the sensorless chain corresponds point by point with the Initial Sensorless Approach.

Regarding the generalization of the proposed Improved Sensorless Approach, it is important to note that the method is strictly developed based on the electrical and dynamic model of the Brushless DC (BLDC) motor. The rotor position extraction logic is intrinsically linked to the trapezoidal profile of the back-EMF, specifically leveraging its characteristic waveform to identify commutation points. Consequently, while the Disturbance Observer (DOB) framework itself is a universal control tool, the sensorless algorithm presented in this study is dedicated to machines with trapezoidal back-EMF. For application to other motor types, such as the Permanent Magnet Synchronous Motor (PMSM) which exhibits a sinusoidal back-EMF, the position estimation logic would require a fundamental adaptation, notably through the use of reference frame transformations (e.g., Park and Clarke) and different signal processing techniques.

Although the proposed Improved Sensorless Approach was validated in a continuous-time simulation environment, its computational requirements for real-time implementation remain low. The Standard Disturbance Observer (SDO) is characterized by a linear, low-order state-space structure, which avoids the use of computationally intensive operations such as high-dimensional matrix inversions, extended Kalman filter (EKF) recursions, or complex trigonometric functions. The reconstruction of the back-EMF involves only basic arithmetic operations (multiplication and addition) and a first-order integration process. Consequently, the algorithm is highly suitable for execution on standard industrial digital signal processors (DSPs) or microcontrollers, even at high switching frequencies (e.g., 20 kHz), where the execution time is expected to be well within the available sampling period.

The proposed Improved Sensorless Approach estimates the back-electromotive force (back-EMF) using the measured phase voltages and currents of the BLDC motor. In this formulation, the derivative of the phase currents is obtained through a disturbance observer rather than direct numerical differentiation, thereby avoiding noise amplification and phase delay issues. However, since the back-EMF is proportional to the rotor speed, it is inherently undetectable at standstill, making any back-EMF-based algorithm ineffective for starting the motor.

To address this limitation, the starting procedure proposed in [20] was implemented. This method involves aligning the rotor to a known position, followed by open-loop acceleration using a frequency ramp. The minimum speed maintained during the simulations is 100 rpm, ensuring a sufficient back-EMF level for the observer to converge. For the sake of brevity, this starting procedure is not detailed here, as the primary objective of this paper focuses on the observer's performance and robustness within its nominal operating range.

4. RESULTS AND DISCUSSION

To evaluate the performance and robustness of the proposed sensorless method, four distinct scenarios are investigated and modeled in MATLAB/Simulink. First, a comparative study is conducted between the proposed approach and an existing method from the literature to establish a performance benchmark. Subsequently, the analysis is extended to

This model is composed of five main blocks:

- **BLDC_Motor:** This is the block diagram of the BLDC motor which has as inputs the v_a , v_b and v_c voltages supplied by the inverter, and the load torque;
- **Inverter:** The block diagram of the inverter, which has as input the switching signals and the voltage of the DC source;
- **Control_System:** It contains a PI speed controller and a PWM signal generator;
- **Initial_sensorless_method:** It contains the implementation of the Initial Sensorless Approach. Its inputs are the voltages V_a , V_b , and V_c , as well as the phase currents i_a , i_b , and i_c . It provides estimates of rotor speed and position as a virtual Hall sensor signal;
- **Improved_sensorless_method:** It implements the Improved Sensorless Approach. Its inputs are the voltages V_a , V_b , and V_c , as well as the phase currents i_a , i_b , and i_c . It provides estimates of rotor speed and position as a virtual Hall sensor signal.

The **Scope_Measure** and **Comparison** blocks are used to visualize the different signals.

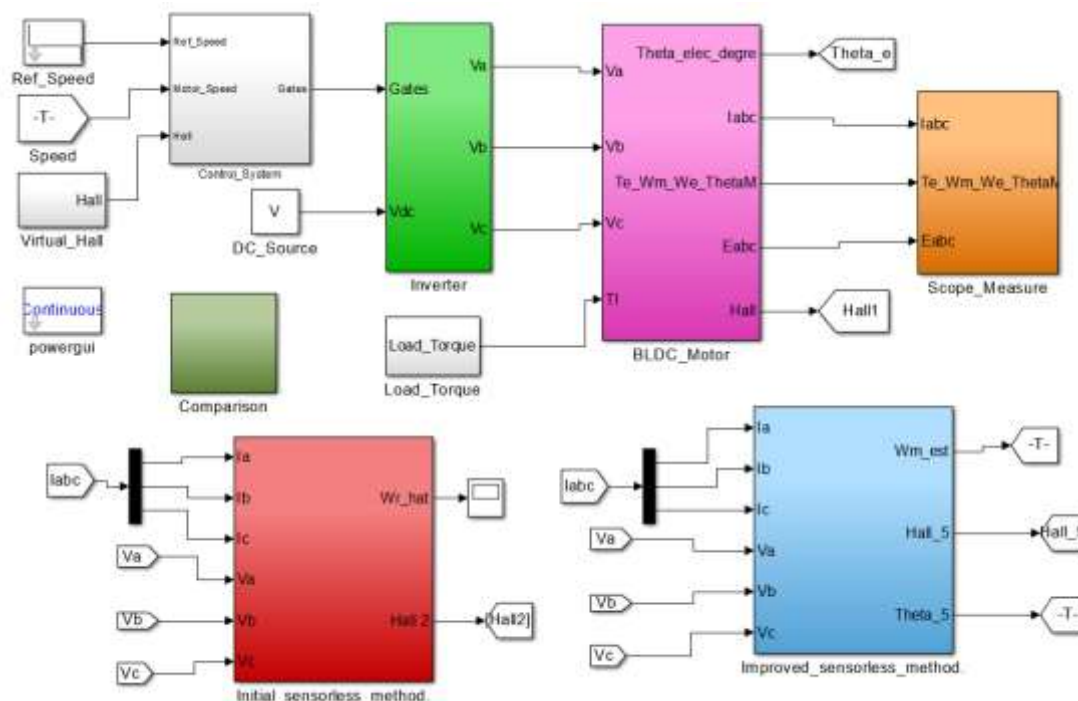


Figure 9: Global Simulink/Matlab model of the proposed BLDC motor drive system used for simulation and performance evaluation. The model integrates the BLDC motor, the inverter, the control system, and both the Initial and Improved Sensorless Approaches within a unified simulation framework. Additional measurement and comparison blocks are included to monitor system variables and directly compare simulation results via dedicated scope-based graphical outputs.

Four simulation scenarios are considered to evaluate the performance and robustness of the proposed sensorless control method for the BLDC motor.

The first scenario examines open-loop operation to directly analyze the accuracy of back-EMF reconstruction and the impact of estimation errors on motor performance. The second scenario introduces closed-loop speed control, where the estimated speed and position are fed back into the regulation and commutation loops to assess the method's effectiveness under dynamic conditions.

To further refine and validate the observer, two additional studies are conducted. The third scenario focuses on observer gain tuning, where the bandwidth ratio n (relating observer speed to electrical speed) and the pole separation factor k (defining the placement of p_1 and p_2) are varied to identify the optimal configuration. Finally, the fourth scenario performs a parameter sensitivity analysis using the optimal gains identified previously (with the observer speed fixed at twenty times

the electrical frequency and $p_2 = 10p_1$). This scenario evaluates the impact of stator resistance and inductance variations, defining the operational boundaries of the approach under practical conditions such as thermal drift or magnetic saturation.

The modeled motor is a 100W-rated power motor with a rated DC voltage of 24V and a rated speed of 3600 rpm. Table 3 summarizes the parameter values of the motor model under study.

To ensure a fair and rigorous comparison with existing methods, the proposed observer dynamics were calibrated with a high bandwidth. Specifically, the observer speed ω_o was fixed at twenty times the motor's electrical frequency ω_e ($\omega_o = 20\omega_e$). Regarding pole placement, a non-identical strategy was employed where $p_1 = \omega_o$ and $p_2 = 10\omega_o$. This specific configuration ensures ultra-fast convergence and optimal damping, thereby highlighting the method's superior dynamic tracking capabilities while simultaneously exposing its sensitivity thresholds when subjected to significant parameter mismatches.

Table 3: BLDC motor and control system parameters used for simulation studies. The motor is rated at 100 W with a nominal speed of 3600 rpm and operates from a 24 V DC supply. The selected electrical and control parameters are summarized.

Designation	Value
Nominal power	100W
DC voltage	24V
Stator resistance per phase R_s	0.1296 Ohm
Stator inductance per phase L	1.2960e-4H
Mutual inductance M	6.48e-5 H
Moment of inertia J	1.5e-4 kg m ²
Friction coefficient B	1e-4 Nm/(rad/s)
Load torque T_l	0.2387 Nm
Magnitude of flux linkage λ_p	0.02578 V/rad/sec
Number of pole pairs p	2
Nominal Speed	3600 rpm
Proportional gain K_p (Speed PI controller)	0.0028 V/rpm
Integral gain K_i (Speed PI controller)	0.1500 V.s/rpm
Electrical time constant	1ms
α	1
Inverter Switching Frequency	16 kHz
Simulation Solver: Variable-step (ode45)	

4.1. Scenario 1: Sensorless Control Performance without Speed Regulation

In this scenario, no speed controller is used. The objective is to evaluate the performance of two sensorless methods for estimating rotor speed and rotor position. The two methods are used alternately to control the motor. The position estimated by each method is directly fed into the control law. The first method is designated as the Initial Sensorless Approach. The second method is designated as the Improved Sensorless Approach. Three signals are analyzed: the back electromotive force, the rotor speed, and the rotor position. These signals are compared with those obtained when the motor is controlled using Hall-effect sensors.

1) Comparison of estimated and actual back-emf without speed regulation

This subsection focuses on estimating the back-EMF during open-loop operation. Throughout this phase, the motor is driven under the control of Hall-effect sensors, which serve to establish the reference back-EMF profile. Concurrently, while the motor is running under this sensor-based control, the back-EMF is estimated using both the Initial Sensorless Approach and the Improved Sensorless Approach. This setup enables a direct performance comparison between the actual reference back-EMF signal from the motor and the estimated signals generated by each sensorless method.

Figure 10 shows the noise caused by the derivative of the current measured on the Back-EMF of the Initial Sensorless Approach. Direct differentiation of the measured phase currents introduces significant noise in the unfiltered estimated back-EMF, with peak amplitudes exceeding twice the actual back-EMF magnitude. Low-pass filtering effectively attenuates these noise spikes but introduces a phase delay relative to the actual back-EMF, which adversely affects the subsequent speed and position estimation. This figure shows the actual back-EMF E_a of the motor (phase a). $E_{a_hat_noFilter}$ denotes the estimated back-EMF without filtering. E_{a_hat} represents the estimated and filtered back-EMF.

Furthermore, using a filter can have disadvantages. It can introduce a slight delay between the motor's actual back EMF and the estimate from this method. This delay will be reflected in the estimated position, leading to a loss of motor performance. Indeed, this delay means that, when controlling the motor using the estimated position, under the same operating conditions as with Hall effect sensors, the motor speed decreases. To achieve the same speed as with Hall-effect sensors, it will be necessary either to increase the supply voltage or to reduce the load torque.

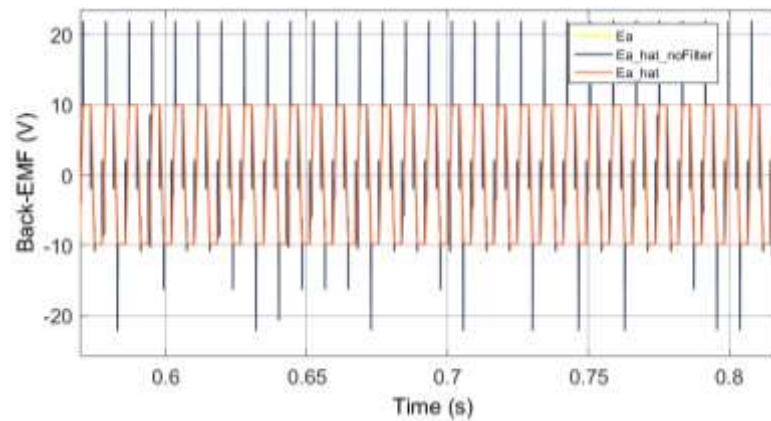


Figure 10: Estimated back-EMF obtained using the Initial Sensorless Approach and comparison with the actual motor back-EMF

A closer look at Figure 11 clearly highlights the high-amplitude noise spikes caused by direct current differentiation and the time shift introduced by low-pass filtering. The observed peaks and phase delay relative to the actual back-EMF explain the degradation in speed and position estimation.

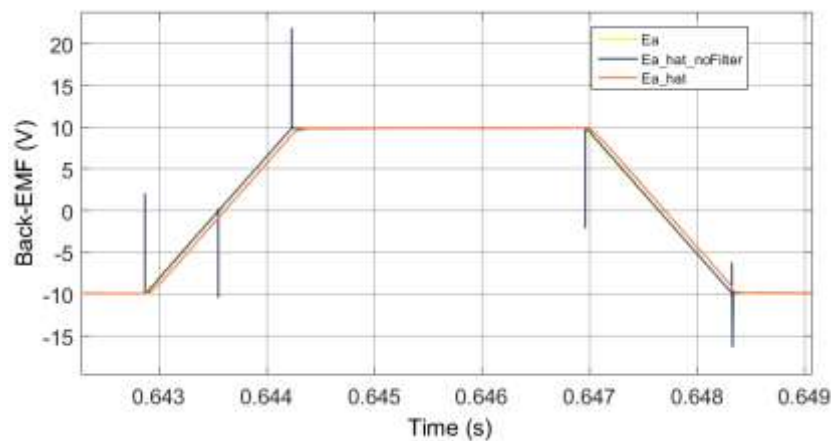


Figure 11: Zoomed view of one half-period of the back-EMF waveforms estimated using the Initial Sensorless Approach.

By using a state observer that provides the derivative of the current, there is no need for a filter on the estimated back-EMF, and therefore, no loss of motor performance.

Figure 12 illustrates the comparison between the back-EMF of the Improved Sensorless Approach Ea_{hat} and actual back-EMF Ea . The observer-based current derivative estimation eliminates the noise spikes observed in the initial approach and removes the phase delay introduced by filtering. As a result, the estimated back-EMF closely matches the actual waveform in both amplitude and phase.

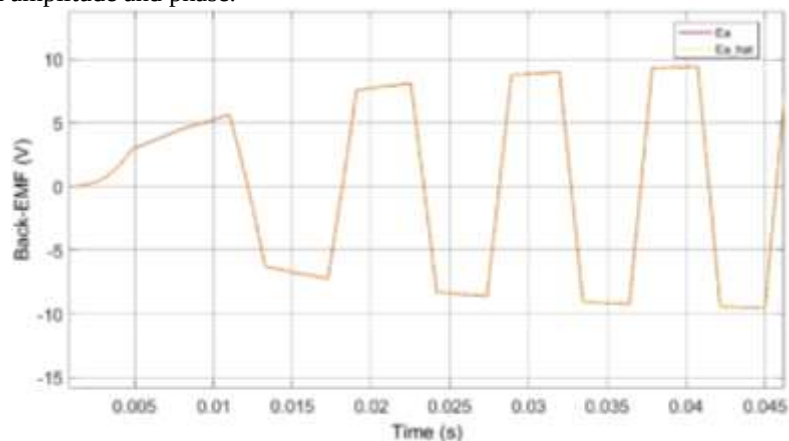


Figure 12: Estimated back-EMF obtained using the Improved Sensorless Approach and comparison with the actual motor back-EMF.

2) Comparison of estimated and measured rotor speed

This subsection focuses on the analysis of the rotor speed response. The objective is to evaluate the impact of the sensorless method on the motor speed in open-loop operation. The speed obtained under Hall sensor-based control is used as a reference. The motor is then driven alternately by the *Initial Sensorless Approach* and the *Improved Sensorless Approach*. The estimated speed provided by each method is analyzed. A comparison between the three speed signals is performed to assess the control performance in the absence of speed regulation.

Figure 13 compares three rotor speed signals. The first signal is the actual motor speed under Hall sensor-based control (Motor Speed). The second signal is the speed estimated by the *Initial Sensorless Approach* when it is used to drive the motor (Speed Sensorless 1). The third signal is the speed estimated by the *Improved Sensorless Approach* when it is used to drive the motor (Speed Sensorless 2). The speed achieved with the Initial Sensorless Approach is significantly lower than that obtained with mechanical Hall-effect sensors, indicating a degradation in motor performance.

In contrast, the Improved Sensorless Approach accurately reproduces the speed obtained with Hall sensors, demonstrating equivalent steady-state performance. These results highlight the impact of the sensorless estimation quality on the open-loop speed performance.

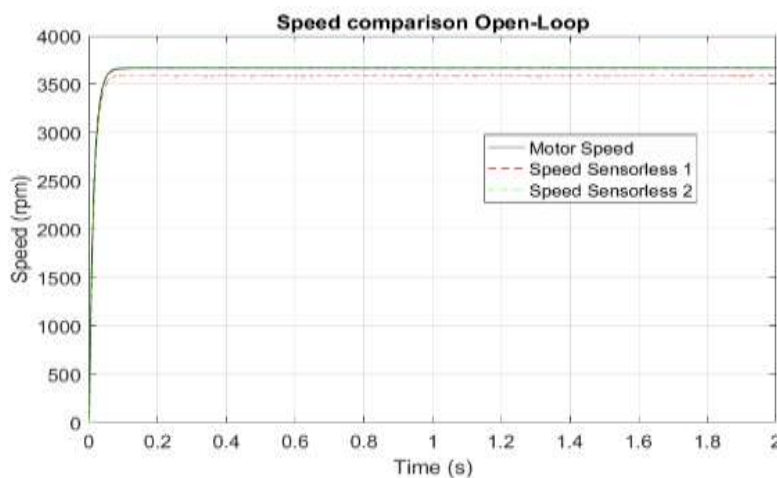


Figure 13: Comparison of motor speed responses obtained without a speed controller using different rotor state acquisition methods.

A closer look at Figure 14 shows the motor speed in steady state. When controlled by Hall effect sensors, the RMS speed is 3666 rpm. The *Initial Sensorless Approach* gives 3588 rpm. The *Improved Sensorless Approach* gives 3669 rpm. The speed error is 0.082% for the Improved Sensorless Approach. It is 2.13% for the Initial Sensorless Approach.

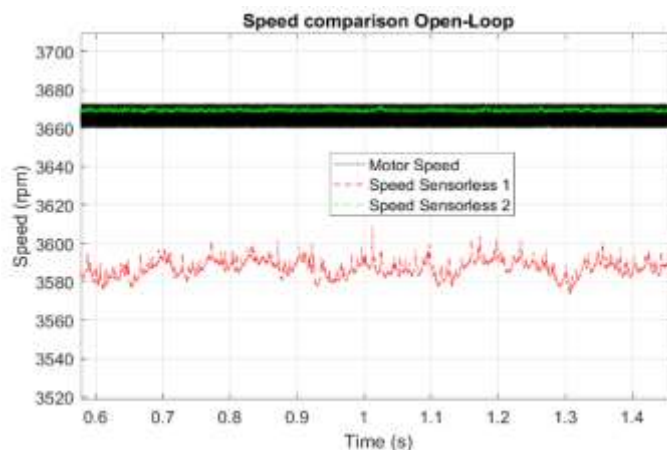


Figure 14: Enlarged view of the open-loop speed comparison. The Initial Sensorless Approach exhibits a clear speed reduction, whereas the Improved Sensorless Approach matches the Hall-effect sensor-based speed, confirming its superior performance.

Table 4 presents a comparison of motor speed for different load conditions. The table compares the motor speed obtained with Hall-effect sensors, the Initial Sensorless Approach, and the Improved Sensorless Approach for load levels of 0%, 50%, 100%, and 150% of nominal torque of 0.2387Nm. For each load, the rotor speed is measured under Hall sensor-based control (Motor Speed). It is also estimated using the *Initial Sensorless Approach* (Estimated Speed 1) and the *Improved Sensorless Approach* (Estimated Speed 2). A clear performance degradation is observed with the Initial

Sensorless Approach, with speed errors of approximately 2%, whereas the improved approach limits the steady-state speed error to less than 0.085%.

Table 4: Steady-state motor speed comparison under different load conditions using sensor-based and sensorless control strategies.

Test	Load	Motor Speed (rpm)	Estimated Speed 1 (rpm)	Error 1 (%)	Estimated Speed 2 (rpm)	Error 2 (%)
1	100%	3666	3588	2,128	3669	0.082
2	50%	3779	3694	2,249	3776	0.079
3	0%	3893	3803	2,312	3896	0.077
4	150%	3556	3482	2,081	3559	0.084

3) Comparison of estimated and real rotor position

This subsection focuses on the rotor position estimation in open-loop operation. The motor is driven by Hall-effect sensors that provide the reference position. The rotor position is also estimated using the *Initial Sensorless Approach* and the *Improved Sensorless Approach*.

A comparison of these three position signals is performed to evaluate the accuracy of each sensorless method. This analysis highlights the performance differences between the two methods in estimating rotor position without speed regulation.

Figure 15 shows the estimation of rotor position. Where the correspondence between the real motor position (in electrical degrees) and the estimated position obtained with the Initial Sensorless Approach and the Improved Sensorless Approach is observed. *Theta_m* represents the rotor's actual position. *Theta1* denotes the position estimated by the Initial Sensorless Approach. *Theta_2* is the position estimated by the Improved Sensorless Approach. Position estimation errors are observed with the Initial Sensorless Approach. The improved approach achieves a rotor position almost identical to that obtained with mechanical sensors.

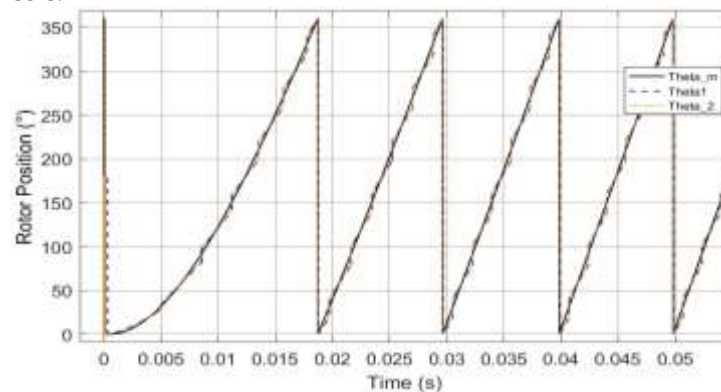


Figure 15: Rotor position comparison using Hall-effect sensors, the Initial Sensorless Approach, and the Improved Sensorless Approach.

Figure 16 shows that the Hall-effect signal of position estimated by Initial Sensorless Approach *Hall_1* lags behind the actual motor position. Whereas the position estimated by the Improved Sensorless Approach *Hall_2* perfectly matches the actual motor position *Hall_mot*.

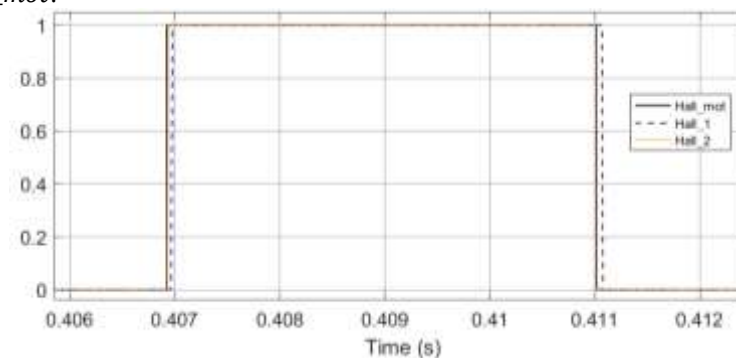


Figure 16: Reconstructed Hall-effect signals obtained from sensorless rotor position estimation. The Initial Sensorless Approach produces delayed commutation signals due to phase lag in the filtered back-EMF. The Improved Sensorless Approach eliminates this delay and provides Hall-equivalent signals synchronized with the sensor-based reference.

4.2. Scenario 2: Sensorless Control Performance with Speed Regulation

In this scenario, a speed controller is included in the control loop. The motor is driven using sensorless control methods, applied alternately. For each method, the estimated speed and rotor position are used for motor control. The estimated speed is fed to the speed controller and compared with the reference speed. The estimated rotor position is used by the control system to generate the commutation signals. The analysis mainly focuses on the speed response. Three speed signals are observed and compared with the reference. The first signal is the actual motor speed. The second signal is the speed estimate from the *Initial Sensorless Approach*. The third signal is the speed estimate from the *Improved Sensorless Approach*.

Figure 17 shows the Simulink implementation of the speed control loop for the BLDC motor. It consists of a Speed controller and a PWM signal generator. The speed controller regulates the motor speed according to a reference signal and drives a PWM signal generator. The generated PWM signals are used to command the inverter that powers the BLDC motor.

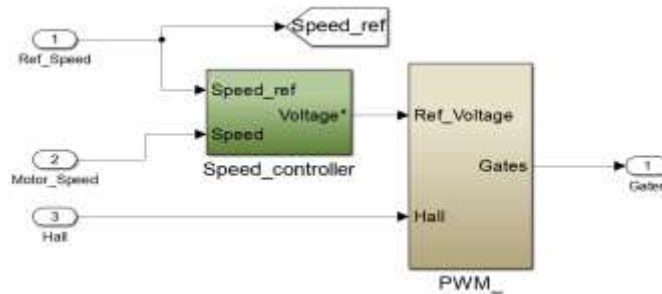


Figure 17: Simulink implementation of the speed control loop for the BLDC motor.

The operating conditions are the same for both sensorless methods: DC voltage, load, and speed controller gains. In case 1, as shown in Figure 18, the rotor speed and position estimated by the Initial Sensorless Approach are used to control the motor. *Real Speed* is the actual speed of the motor. *Reference Speed* is the speed setpoint, and *Speed Sensorless 1* represents the speed estimated by the Initial Sensorless Approach. The speed controller tracks the reference signal based on the Initial Sensorless Approach. While the estimated speed closely follows the reference, the actual motor speed slightly exceeds the reference at low-speed setpoints but matches it well at high speeds. Figure 18 shows that when using the speed estimated by the Initial Sensorless Approach in a speed controller, the real motor speed is slightly higher than the setpoint at low speeds.

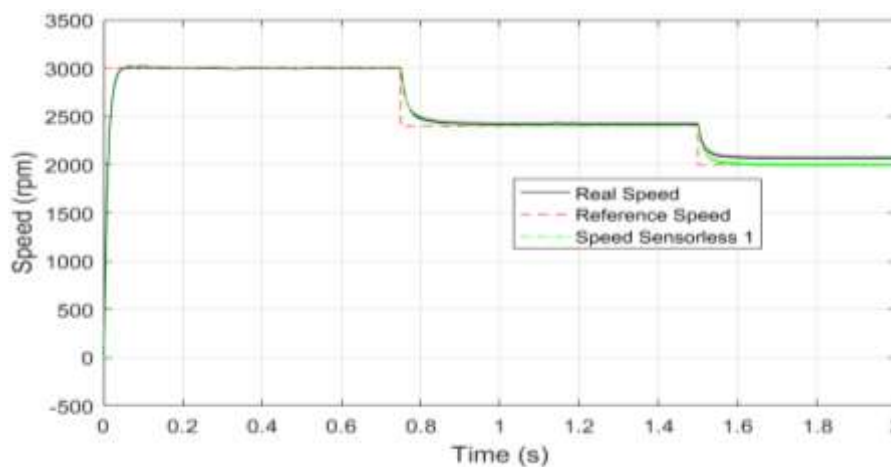


Figure 18: Closed-loop speed comparison using the estimated rotor speed for feedback.

Table 5 shows the comparison of the RMS values of these different speeds in steady state. Larger deviations between the actual motor speed and reference occur at low-speed setpoints, whereas the estimated speed closely follows the reference across all conditions.

Table 5: Comparison of reference, actual, and estimated motor speeds for different setpoints in closed-loop control.

Test	Ref Speed (rpm)	Motor Speed (rpm)	Estimated Speed 1 (rpm)	Error (%)
1	3000	3003	3000	0.100
2	2400	2425	2400	1.042
3	2000	2071	2001	3.550

In case 2, as shown in Figure 19, the rotor speed and rotor position estimated by the Improved Sensorless Approach are used to control the motor. *Real Speed* represents the motor's actual speed. *Reference Speed* denotes the speed setpoint. *Speed Sensorless 2* is the speed estimated by the Improved Sensorless Approach. The estimated speed is used to provide feedback to the controller. The three speeds are closely aligned across all tested reference values, demonstrating the Improved Sensorless Approach's higher accuracy compared with the Initial Sensorless Approach.

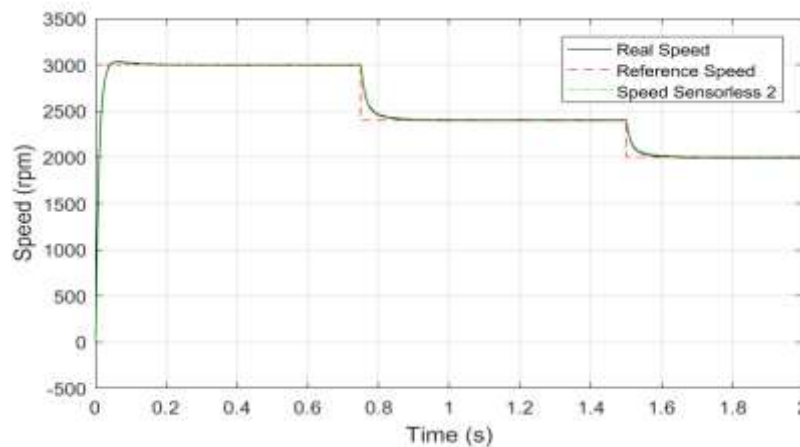


Figure 19: Comparison of reference, actual, and estimated motor speeds under closed-loop speed control using the Improved Sensorless Approach.

Table 6 shows the comparison of the RMS values of these different speeds in steady state. The Improved Sensorless Approach maintains very small steady-state speed errors across all reference speeds, particularly at low setpoints, outperforming the Initial Sensorless Approach.

Table 6: Comparison of reference, actual, and estimated motor speeds for multiple setpoints under closed-loop control

Test	Ref Speed (rpm)	Motor Speed (rpm)	Estimated Speed 2 (rpm)	Error (%)
1	3000	2999	3000	0.033
2	2400	2400	2400	0.000
3	2000	1998	2000	0.100

The residual speed error (e.g., 0.033% in Table 6) originates from a combination of observer convergence dynamics, solver tolerances, and minor waveform distortions during phase commutations.

4.3 Scenario 3: Influence of the Observer Gain Selection on the Performance of the Proposed Method

In Scenario 3, the influence of the observer gain selection on the performance of the proposed sensorless method is investigated. The observer gains are tuned through the observer bandwidth ω_o , defined as a multiple of the motor electrical angular frequency ω_e . Several values are considered according to $\omega_o = n\omega_e$, with $n = 5, 10, 15, 20, 25$ and 30 , in order to evaluate the effect of increasing the observer dynamics on the estimation accuracy.

The observer tuning is directly informed by the motor's physical characteristics. Given an electrical time constant of 1 ms and a nominal electrical speed of approximately $\omega_e = 1000 \text{ rad/s}$, the observer dynamics were calibrated to ensure rapid convergence. To provide additional flexibility in the gain tuning, the observer poles are selected as $p_1 = \omega_o$ and $p_2 = k\omega_o$, where k is a positive real constant. The simulations are first performed with identical poles $k = 1$, then repeated for non-identical pole placement with $k = 5$, $k = 10$, and $k = 20$. For each configuration, the rotor speed response is presented and the estimation performance is quantified using the Mean Absolute Error (MAE) of the estimated speed. In addition, bar charts are used to summarize the average MAE values for the speed estimation and the average Root Mean Square Error (RMSE) values for the rotor position estimation, allowing a clear comparison between the different observer bandwidth settings and pole placement strategies.

Figure 20 presents the estimated rotor speed obtained for different observer gain settings under identical pole placement. The estimated speeds are compared with the actual motor speed measured using Hall-effect sensors. It can be observed that increasing the observer bandwidth significantly improves the speed estimation accuracy, as the estimated speed progressively converges toward the real rotor speed. The results indicate that the proposed method provides excellent performance starting from $n = 10$, where the estimated speed becomes very close to the measured speed.

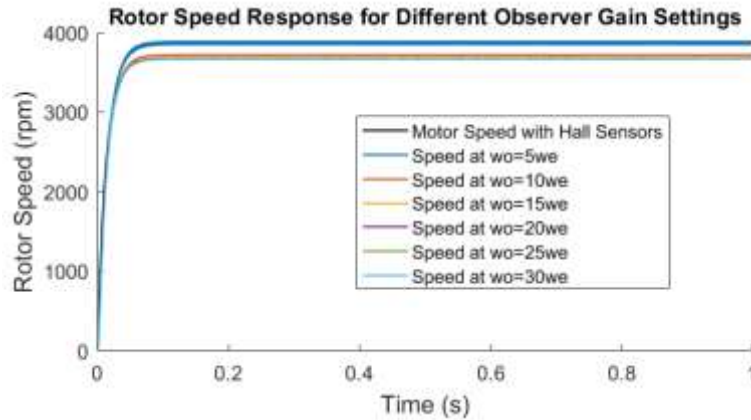


Figure 20: Rotor speed response for different observer gains (identical observer poles)

Figure 21 shows the Mean Absolute Error (MAE) of the rotor speed estimation as a function of the observer gain settings under identical pole placement. This figure confirms the trend observed in Figure 20: as the observer bandwidth increases, the MAE decreases significantly, indicating an improvement in estimation accuracy. The error becomes very small for $n \geq 10$, demonstrating that the proposed method achieves reliable speed estimation when the observer is tuned with a sufficiently high bandwidth.

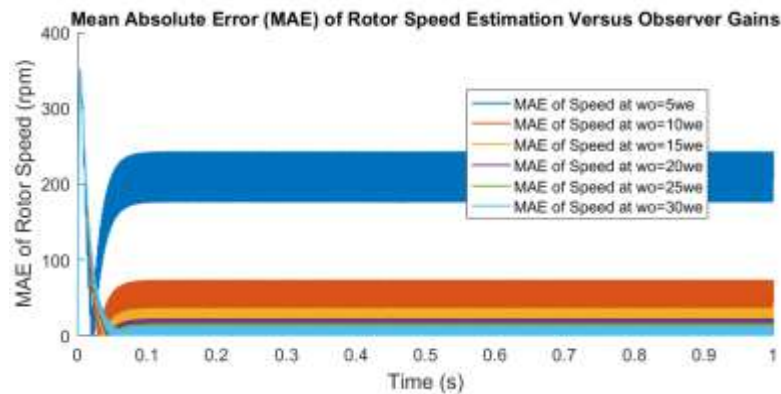


Figure 21: Mean absolute error (MAE) of rotor speed estimation versus observer gains (identical observer poles)

Figure 22 illustrates the average Mean Absolute Error (MAE) of the rotor speed estimation for different observer bandwidth ratios. At $n = 5$, the average MAE reaches 5.81%, indicating insufficient observer dynamics. When the bandwidth is increased to $n = 10$, the MAE is significantly reduced to 1.52%, and continues to decrease for higher values of n . These results clearly confirm that increasing the observer bandwidth improves the speed estimation accuracy and enhances the overall performance of the proposed sensorless method.

Figure 23 shows the average Mean Absolute Error (MAE) of the rotor speed estimation for different observer bandwidth ratios when using non-identical pole placement with $k = 5$. Compared to the identical pole configuration, the results demonstrate that assigning two distinct observer poles significantly improves the estimation accuracy. By selecting a faster second pole ($p_2 > p_1$), the observer dynamics become more responsive, leading to lower speed estimation errors and improved rotor position estimation performance. This confirms that non-identical pole placement provides a more effective gain tuning strategy for the proposed sensorless method.

Figure 24 presents the average Mean Absolute Error (MAE) of the rotor speed estimation for different observer bandwidth ratios using non-identical pole placement with $k = 10$. The results show that increasing the pole separation further improves the speed estimation accuracy, leading to a lower MAE compared to the case $k = 5$. However, for larger values of n , a slight increase in the MAE can be observed. This behavior may be attributed to the higher observer bandwidth, which increases sensitivity to measurement noise and inverter switching disturbances, thereby affecting the estimation performance at very high gain levels.

Figure 25 illustrates the average Mean Absolute Error (MAE) of the rotor speed estimation for different observer bandwidth ratios using non-identical pole placement with $k = 20$. Unlike the cases $k = 5$ and $k = 10$, the results show that the MAE tends to increase as n becomes larger. This indicates that excessively separating the observer poles leads to overly aggressive observer dynamics, which reduces robustness and increases sensitivity to noise and switching effects. Consequently, the configuration $k = 20$ becomes less effective and provides lower performance compared to the optimal case $k = 10$.

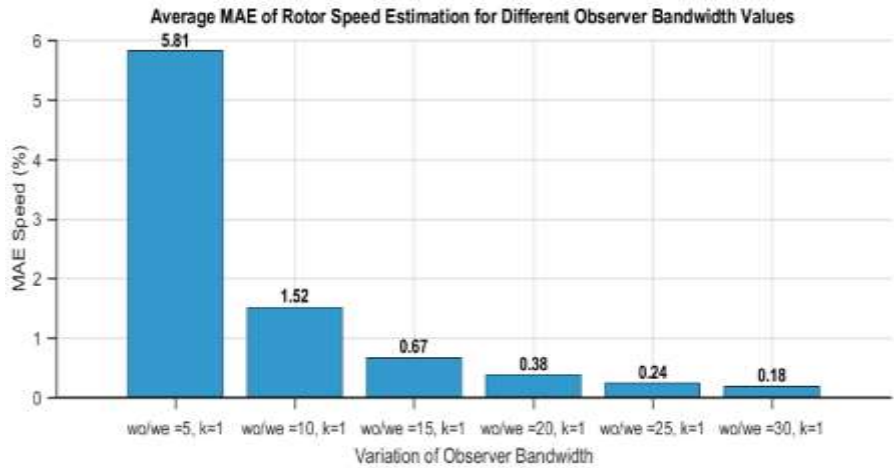


Figure 22: Average MAE of rotor speed estimation for different observer bandwidth values, $k=1$

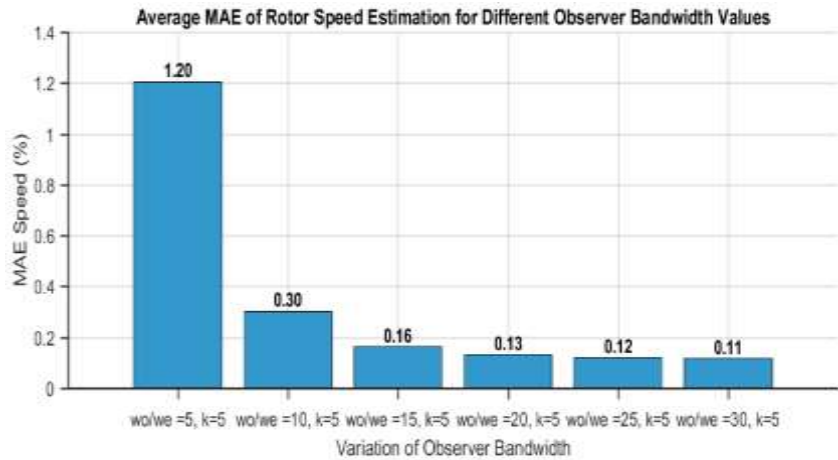


Figure 23: Average MAE of rotor speed estimation for different observer bandwidth values, $k=5$

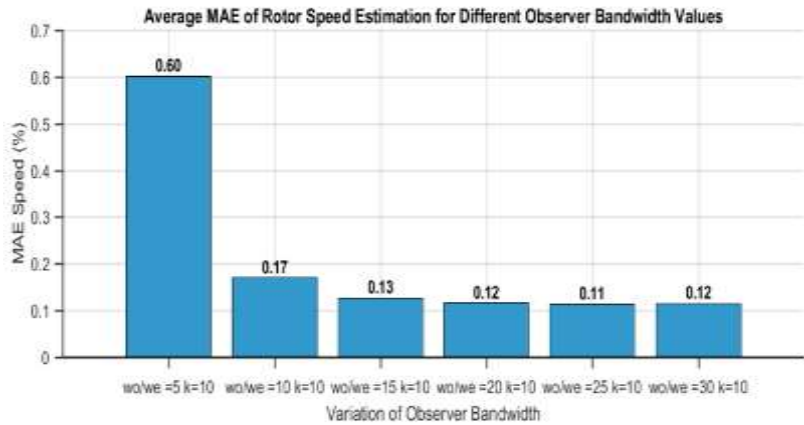


Figure 24: Average MAE of rotor speed estimation for different observer bandwidth values, $k=10$

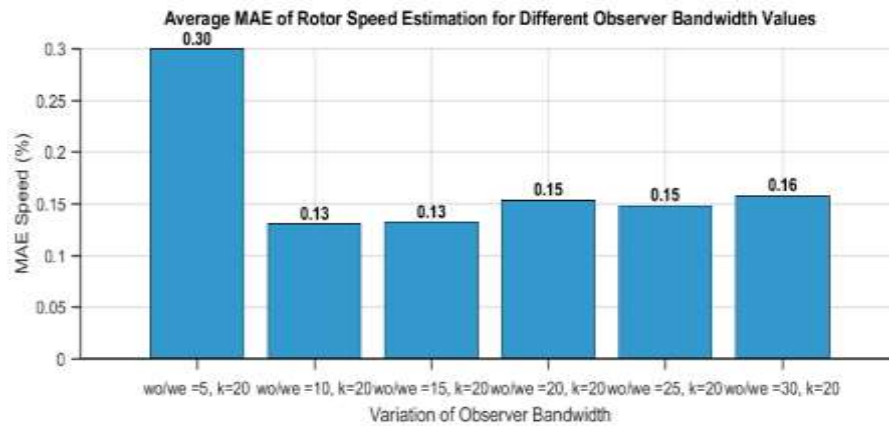


Figure 25: Average MAE of Rotor Speed Estimation for different Observer Bandwidth values, $k=20$

Figure 26 presents the average Root Mean Square Error (RMSE) of the rotor position estimation for different observer bandwidth ratios. The RMSE is 5.66° for $n = 5$ and slightly decreases to 5.58° at $n = 10$. It then increases marginally to 5.65° at $n = 15$, and shows a slight upward trend for higher bandwidth values. This behavior indicates that increasing the observer bandwidth significantly improves speed estimation accuracy, while the position estimation RMSE remains almost constant and may slightly degrade at very high bandwidth settings due to increased sensitivity to noise and switching disturbances.

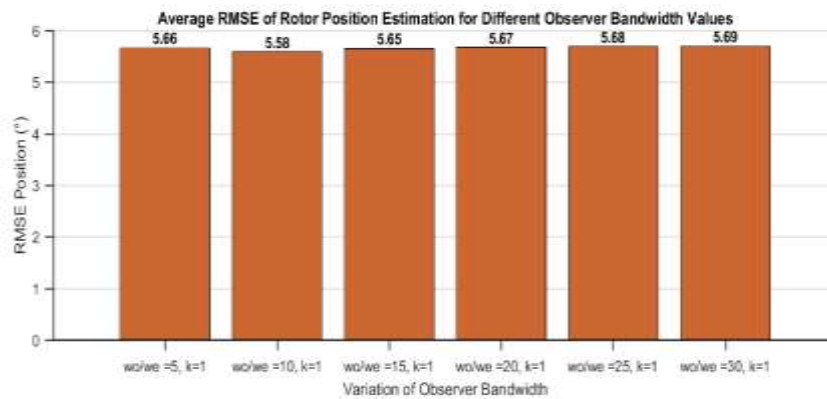


Figure 26: Average RMSE of rotor position estimation for different observer bandwidth values, $k=1$

Figure 27 presents the average Root Mean Square Error (RMSE) of the rotor position estimation for different observer bandwidth ratios using non-identical pole placement with $k = 5$. It can be observed that the configuration $n = 5$ provides the lowest position RMSE. As n increases, the RMSE slightly increases, although the variation remains very small. The error stays almost constant for bandwidth ratios between $n = 15$ and $n = 25$, indicating stable position estimation performance over this range. However, at $n = 30$, a noticeable increase in RMSE is observed, which may be explained by the higher sensitivity of the observer to noise and switching disturbances at very large bandwidth values.

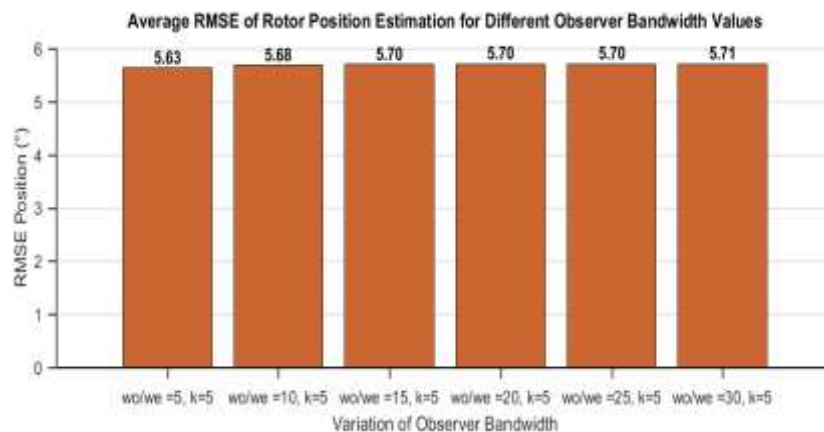


Figure 27: Average RMSE of rotor position estimation for different Observer bandwidth values, $k=5$

Figure 28 and Figure 29 show the average Root Mean Square Error (RMSE) of the rotor position estimation for different observer bandwidth ratios using non-identical pole placement with $k = 10$ and $k = 20$, respectively. The results indicate that increasing the observer bandwidth has only a limited effect on the position estimation accuracy. Overall, the RMSE values remain nearly constant over the considered range of n , showing that the improvement in rotor position estimation is relatively small compared to the speed estimation enhancement. For instance, in the case $k = 10$, the RMSE remains almost identical for $n = 5$ and $n = 30$, confirming that the impact of higher observer gains on the position estimation performance is weak.

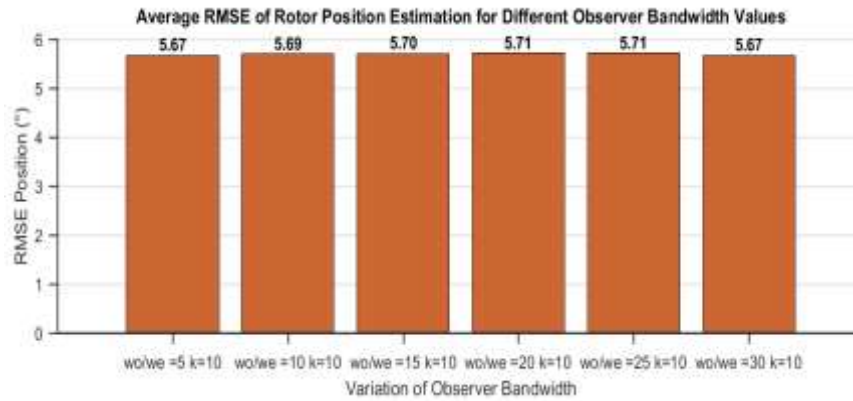


Figure 28: Average RMSE of rotor position estimation for different observer bandwidth values, $k=10$

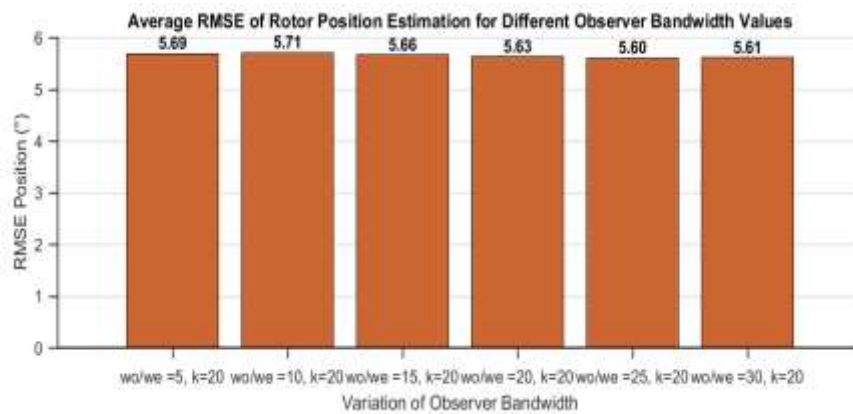


Figure 29: Average RMSE of rotor position estimation for different observer bandwidth values, $k=20$

The results of Scenario 3 clearly demonstrate that the observer gain selection has a significant impact on the performance of the proposed sensorless method, particularly in terms of speed estimation accuracy. When the observer bandwidth ratio $\frac{\omega_o}{\omega_e} = n$ increases, the Mean Absolute Error (MAE) of the estimated rotor speed decreases considerably, showing that faster observer dynamics improve the reconstruction of the back-EMF and enhance the speed estimation quality. Moreover, using non-identical pole placement ($p_2 = kp_1$) provides additional tuning flexibility and leads to better performance compared to identical poles, with the best results obtained around $k = 10$. However, excessively increasing the pole separation (e.g., $k = 20$) tends to degrade the performance, mainly due to higher sensitivity to noise and switching disturbances. In contrast, the Root Mean Square Error (RMSE) of the rotor position estimation remains almost constant for all tested gain configurations, indicating that the observer bandwidth has only a limited influence on the position estimation accuracy. Overall, these results confirm that an appropriate gain tuning strategy is essential to achieve accurate speed estimation while maintaining stable and consistent rotor position estimation performance.

4.4 Scenario 4: Sensitivity Analysis to Motor Parameter Variations (Fixed Observer Gains)

In Scenario 4, a sensitivity analysis is carried out to evaluate the robustness of the proposed sensorless method against uncertainties in the motor electrical parameters. The observer gains are kept constant and are tuned based on the nominal values of the stator resistance and inductance. Three different cases are investigated: Case 1, where only the stator resistance R is varied while L_m remains constant; Case 2, where only the equivalent inductance L_m is varied while R is kept constant; and Case 3, where both parameters R and L_m are varied simultaneously. For each case, the performance of the proposed method is mainly assessed using the average Mean Absolute Error (MAE) of the estimated rotor speed and the average Root Mean Square Error (RMSE) of the estimated rotor position. This scenario provides a clear evaluation of the estimation accuracy degradation under parameter variations and highlights the robustness limits of the proposed sensorless approach.

Figure 30 presents the average Mean Absolute Error (MAE) of the rotor speed estimation for different stator resistance values. The results show that decreasing the stator resistance has only a limited impact on the speed estimation accuracy. For instance, at the nominal resistance value ($R=100\%$), the MAE is 0.12%, and it slightly increases to 0.36% when the resistance is reduced to 80% of its nominal value. In contrast, increasing the stator resistance significantly degrades the estimation performance. The MAE rises to 4.25% at $R=120\%$, 6.68% at $R=130\%$, and reaches 12.71% at $R=150\%$ of the nominal resistance. These results indicate that, to maintain a speed estimation MAE below 5%, the stator resistance should not exceed approximately 120% of its nominal value.

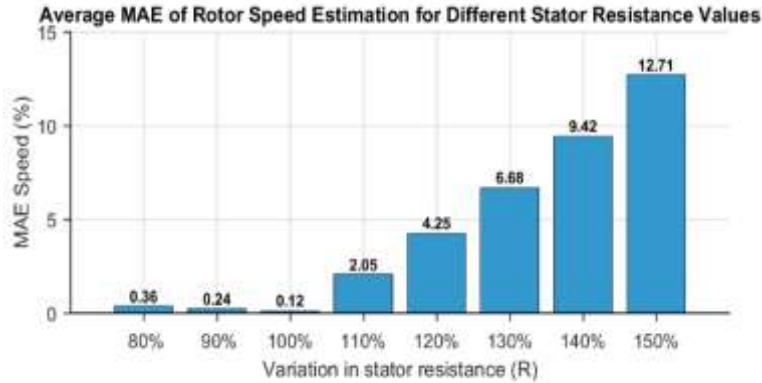


Figure 30: Average MAE of rotor speed estimation for different stator resistance values

Figure 31 shows the average Root Mean Square Error (RMSE) of the rotor position estimation for different stator resistance values. The results indicate that reducing the stator resistance slightly improves the position estimation accuracy, as the RMSE obtained at $R=80\%$ of the nominal value is lower than that at the nominal resistance ($R=100\%$). However, increasing the stator resistance leads to a noticeable degradation in performance. The RMSE reaches 6.19° at $R=110\%$ and increases significantly to 10.91° at $R=150\%$ of the nominal resistance. These results suggest that, in order to maintain the rotor position estimation error around 6° , the stator resistance should not exceed approximately 110% of its nominal value.

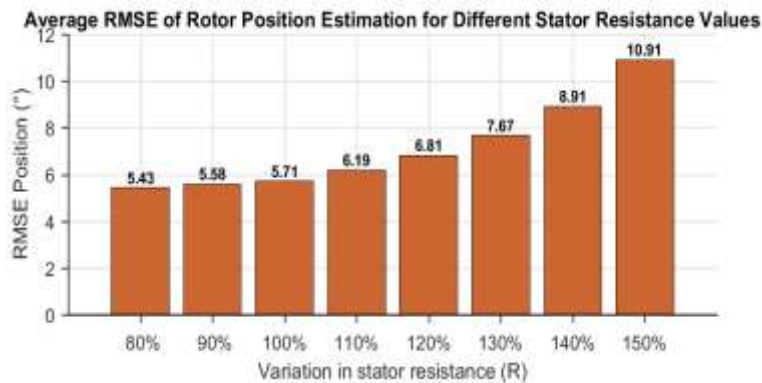


Figure 31: Average Root Mean Square Error of rotor position estimation for different stator resistance values

Figure 32 presents the average Mean Absolute Error (MAE) of the rotor speed estimation for different equivalent inductance values L_m . The results show that variations in L_m affect the estimation accuracy, even when the observer gains are kept constant. When the inductance decreases to 80% of its nominal value, the speed MAE increases to 0.88%, which is higher than the 0.12% obtained under nominal conditions ($R = 100\%$, $L_m = 100\%$). On the other hand, increasing the inductance to 120% of its nominal value leads to a significant degradation of performance, with the MAE reaching approximately 4.41%. These results indicate that the proposed method is moderately sensitive to inductance variations, especially when L_m increases above its nominal value.

Figure 33 illustrates the average Root Mean Square Error (RMSE) of the rotor position estimation for different equivalent inductance values L_m . The results show that decreasing the inductance below its nominal value increases the position estimation error. For example, at $L_m = 90\%$ of the nominal inductance, the RMSE reaches 6.33° , and it further increases to 7.04° when L_m is reduced to 80% of its nominal value. Similarly, an increase in inductance also degrades the estimation accuracy. At $L_m = 120\%$ of the nominal value, the position RMSE rises to 7.69° . Overall, these results indicate that the rotor position estimation accuracy is sensitive to inductance variations, and the RMSE tends to increase when L_m deviates significantly from its nominal value.

Figure 34 presents the average Mean Absolute Error (MAE) of the rotor speed estimation under simultaneous variations of the stator resistance RRR and the equivalent inductance L_m . In this analysis, stator resistance R is increased while equivalent stator inductance L_m is decreased, which represents a realistic operating condition combining thermal

effects and magnetic saturation. The results show that the speed estimation accuracy degrades significantly under combined parameter deviations. For instance, at $R=120\%$ of its nominal value and $L_m = 90\%$ of its nominal value, the MAE reaches 4.42%. The error exceeds 5% starting from $R=130\%$ and $L_m=85\%$, and continues to increase until it reaches 12.74% at $R=150\%$ and $L_m = 80\%$ of their respective nominal values. These results highlight that the proposed sensorless method becomes more sensitive when both parameters vary simultaneously, leading to a considerable deterioration in speed estimation performance.

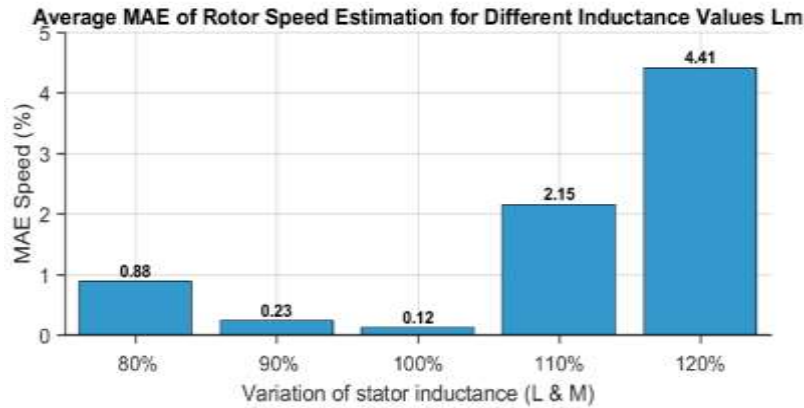


Figure 32: Average Mean Absolute Error of rotor speed estimation for different inductance values

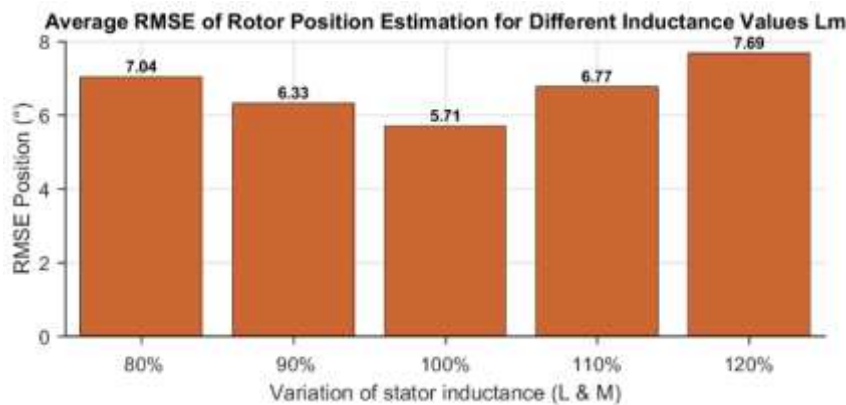


Figure 33: Average Root Mean Square Error of rotor position estimation for different inductance values

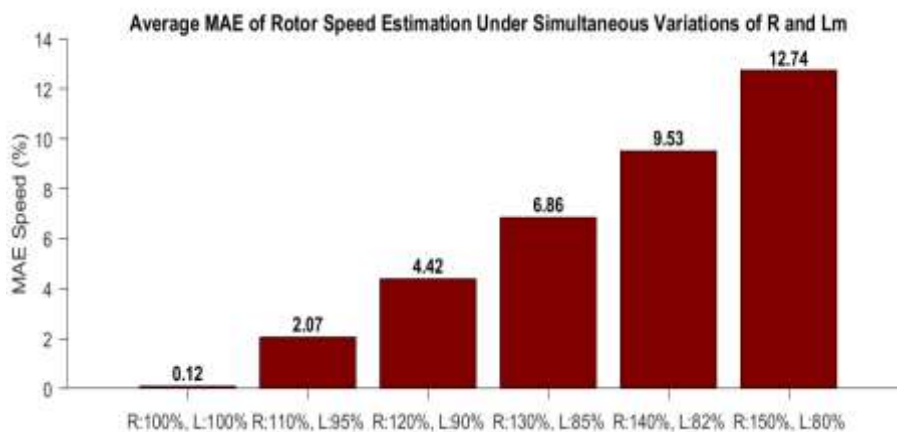


Figure 34: Average Mean Absolute Error of rotor speed estimation under simultaneous variations of stator resistance et stator inductance

Figure 35 shows the average Root Mean Square Error (RMSE) of the rotor position estimation under simultaneous variations of the stator resistance R and the equivalent inductance L_m . The results indicate that the position estimation accuracy is strongly affected when both parameters deviate from their nominal values. The RMSE exceeds 6° starting from the first combined variation case. For instance, at $R=110\%$ of its nominal value and $L_m = 95\%$ of its nominal value, the

position RMSE reaches 6.61° . As the deviation becomes more severe, the estimation error increases significantly, reaching 14.10° at $R=150\%$ and $L_m = 80\%$ of their respective nominal values. These results confirm that simultaneous parameter variations considerably degrade the rotor position estimation performance of the proposed sensorless method.

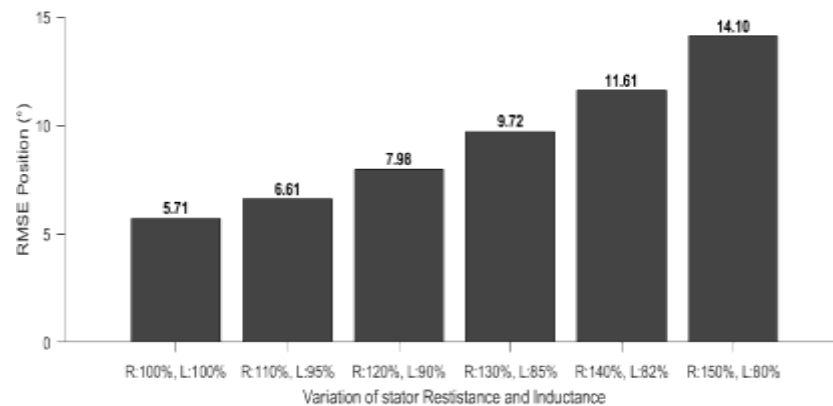


Figure 35: Average RMSE of rotor position estimation under simultaneous variations of stator resistance et stator inductance

The robustness analysis of the proposed sensorless method demonstrates a significant sensitivity to stator parameter mismatches, with resistance R being the most critical factor. While an underestimation of R yields a negligible increase in speed Mean Absolute Error (MAE), an overestimation leads to a non-linear degradation of performance, with the speed MAE reaching 12.71% at 150% of nominal value. Similarly, variations in the equivalent inductance L_m consistently increase both speed MAE and position Root Mean Square Error (RMSE), though the position estimation remains more resilient to L_m fluctuations than to R variations. The most severe performance degradation occurs under simultaneous parameter drifts; specifically, the combination of increased resistance and decreased inductance—often encountered during motor heating and magnetic saturation—compounds the estimation errors, resulting in a maximum position RMSE of 14.10° .

Furthermore, the impact of the parameter α introduced into the rotor position estimation formulas on estimation accuracy was investigated by varying its value from 0.5 to 2. Figure 36 and Figure 37 indicate that the Mean Absolute Error (MAE) for speed remains remarkably stable (approximately 0.12%) within the α range of 0.5 to 1.5. Beyond this threshold, the speed error increases, reaching 1.2% at $\alpha = 2$, which suggests a loss of linearity in the speed reconstruction process.

A much higher sensitivity was observed regarding the Root Mean Square Error (RMSE) of the rotor position. The position error decreases significantly as α increases, reaching an optimal minimum of 2.74° at $\alpha = 1.5$, compared to the initial value at $\alpha = 1$ (RMSE = 5.71°). However, a critical instability threshold is encountered beyond $\alpha = 1.5$, where the RMSE surges to 76.95° at $\alpha = 1.7$ and exceeds 114° at $\alpha = 2$. This indicates a complete loss of synchronization in the position estimation. Conducted with fixed observer gains ($\omega_0 = 20\omega_e$, $p_2 = 10p_1$), this analysis confirms that $\alpha = 1.5$ is the optimal setting to enhance position accuracy while maintaining a negligible speed estimation error.

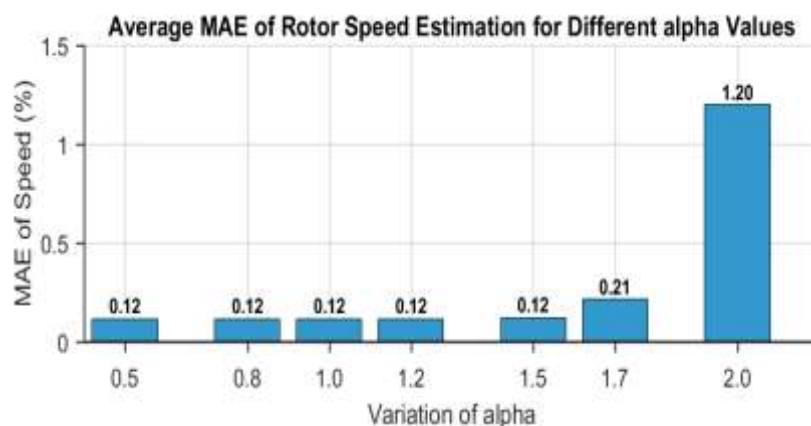


Figure 36: Average Mean Absolute Error of rotor speed estimation for different values of α

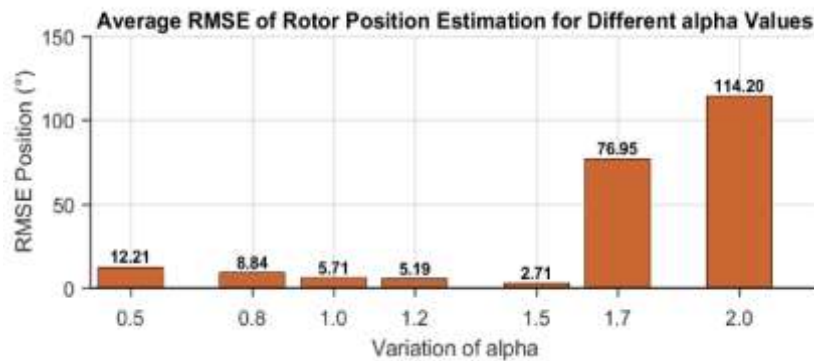


Figure 37: Average Root Mean Square Error of rotor position estimation for different values of α

4.3. Discussion and Limitations

The results highlight the impact of the sensorless estimation method on motor performance. Figures 10 to 12 analyze the quality of back-EMF estimation obtained with the two sensorless approaches. Figure 10 shows that the back-EMF estimated using the Initial Sensorless Approach is significantly affected by high-frequency noise caused by direct numerical differentiation of the measured phase currents. The noise peaks can exceed the actual back-EMF amplitude by a factor of 2, necessitating low-pass filtering. Although the applied filter effectively attenuates these peaks, it introduces a noticeable phase delay between the estimated and actual back-EMF, which directly impacts the accuracy of the derived speed and rotor position.

This phenomenon is further illustrated in the zoomed view of Figure 11, where both the noise peaks and the phase shift over a half electrical period are clearly visible.

In contrast, Figure 12 demonstrates that the Improved Sensorless Approach provides a clean back-EMF estimation without observable noise spikes or phase delay. By estimating the current derivatives through a disturbance observer, the proposed method eliminates the need for filtering and preserves the phase alignment between the estimated and actual back-EMF. These results confirm that the improved back-EMF reconstruction underlies the enhanced speed and position estimation performance observed in subsequent figures.

Figures 13 to 16 collectively highlight the impact of the sensorless strategy on both speed and rotor position estimation. In open-loop operation without speed regulation, Figure 13 shows that the Initial Sensorless Approach results in a noticeable reduction in motor speed compared with Hall-effect sensor-based control, indicating a performance loss. This behavior is further emphasized in the zoomed view of Figure 14, where the speed deficit caused by the Initial Sensorless Approach becomes clearly visible. In contrast, the Improved Sensorless Approach exhibits a speed response that closely matches the Hall-effect sensor reference, confirming that the proposed observer-based formulation effectively preserves nominal motor performance.

The influence of back-EMF estimation accuracy on rotor position reconstruction is illustrated in Figures 15 and 16. While the Initial Sensorless Approach introduces a phase shift in the estimated rotor position, resulting in delayed commutation signals when converted into Hall-equivalent signals, the Improved Sensorless Approach shows excellent agreement with the Hall-effect sensor signals. This improvement is directly attributed to the elimination of phase delay in back-EMF estimation, achieved by replacing direct current differentiation with disturbance-observer-based current derivative estimation. Overall, these results demonstrate that the Improved Sensorless Approach significantly enhances both speed and position estimation accuracy, eliminating the performance degradation observed with the Initial Sensorless Approach.

Figures 18 and 19 present the performance of the sensorless approaches under closed-loop speed control, where the estimated speed is used as the controller's feedback. In Figure 18, when the Initial Sensorless Approach is employed, the estimated speed closely follows the reference, particularly at low setpoints; however, the actual motor speed remains slightly higher than both the reference and the estimated speed. This discrepancy becomes more pronounced at lower speed commands and reflects the impact of the phase delay introduced in the back-EMF estimation, which affects the accuracy of the feedback signal used by the speed controller.

In contrast, Figure 19 demonstrates that the Improved Sensorless Approach achieves excellent closed-loop performance across all tested speed references. The estimated speed, actual motor speed, and reference signal are nearly identical, even at low-speed setpoints. This result confirms that disturbance-observer-based estimation of current derivatives effectively eliminates phase shift and improves back-EMF accuracy, resulting in precise speed feedback and superior control performance. Consequently, the Improved Sensorless Approach enables reliable, accurate speed regulation comparable to that achieved with mechanical sensors, while avoiding the limitations of the Initial Sensorless Approach.

The quantitative results further support this observation. Tables 5 and 6 show that in steady state, the rotor speed estimation error for the Improved Sensorless Approach is approximately 0.1%, while the Initial Sensorless Approach exhibits a significantly higher error of about 3%. These results demonstrate the superior accuracy and reliability of the improved method across different operating conditions.

The robustness analysis of Improved Sensorless Approach was conducted by fixing the observer dynamics to a high-speed configuration ($\omega_o = 20\omega_e$ and $p_2 = 10p_1$) which, according to the gain sensitivity study, minimizes tracking errors under nominal conditions. However, the results demonstrate that even high-performance observer dynamics cannot compensate for the biases introduced by stator parameter mismatches. The speed MAE reaches 12.71% when the resistance is increased to 150% of its nominal value, confirming that the error in the $R \cdot i$ term dominates the back-EMF estimation, rendering the observer's bandwidth secondary to model accuracy.

Regarding rotor position, the resilience of the RMSE to gain adjustments contrasts sharply with its significant degradation during L_m variations. With the position error peaking at 14.10° under simultaneous parameter drifts, it is evident that the observer (despite being well-damped and fast due to the $k=10$ ratio) suffers from a structural phase shift. In the literature, position errors exceeding 10° are typically associated with substantial efficiency losses and increased torque ripple. Consequently, this research highlights that for high-thermal-load applications where R and L_m vary, gain optimization ($n=20$, $k=10$) must be supplemented by parameter compensation mechanisms to maintain performance within acceptable industrial margins.

A detailed assessment of the results allows for the identification of a viable operational envelope for the proposed sensorless method. It is observed that simultaneous parameter variations—specifically with stator resistance R at 120% and equivalent stator inductance L_m at 90% of their nominal values—yield a position RMSE of approximately 8° and a speed MAE below the 5% threshold. Within the context of established literature, these metrics represent an acceptable performance limit for reliable motor control. Consequently, these variation levels can be defined as the maximum tolerance boundaries for the proposed observer when configured with $n=20$ and $k=10$. Beyond these limits, the degradation of the reconstructed back-EMF becomes too significant to ensure optimal commutation without the implementation of additional parameter compensation strategies.

While the proposed sensorless approach demonstrates high estimation precision, certain limitations must be acknowledged. First, although the simulation environment incorporates a 16 kHz switching frequency to approximate real-world inverter behavior, a full experimental validation is required to account for hardware nonlinearities and measurement noise. Second, the motor model assumes constant inductances and neglects magnetic saturation and iron losses; while these simplifications are standard for establishing control frameworks, they may affect accuracy under extreme saturation conditions. Third, while robustness was validated against parameter variations and load disturbances (see Section 4.1, Table 4), the observer remains sensitive to severe model mismatches. Furthermore, the operational range of the sensorless algorithm is focused on speeds above 100 rpm. This threshold represents the handover speed where the control system transitions from the initial open-loop startup strategy to the proposed observer-based closed-loop control, ensuring that the back-EMF signals are sufficiently robust for accurate estimation.

5. CONCLUSION

This paper has presented a sensorless control approach for a brushless DC (BLDC) motor based on back-EMF estimation. The main objective was to develop a simple, computationally efficient method that operates reliably not only at high speeds but also in low-speed regions, where conventional back-EMF-based techniques often face limitations. An initial sensorless approach was investigated, in which the back-EMF is reconstructed from measured phase voltages and currents using the motor's electrical model. However, this method requires low-pass filtering to attenuate noise introduced by direct current differentiation, which causes phase shifts and degraded control performance.

Simulation results showed that the phase delay introduced by filtering in the Initial Sensorless Approach causes a mismatch between the estimated and actual back-EMF, resulting in noticeable performance degradation, particularly in speed and position estimation. To overcome this limitation, an Improved Sensorless Approach was proposed by integrating a disturbance observer to estimate the phase current derivatives. This observer-based formulation eliminates the need for direct differentiation and filtering, thereby suppressing noise and removing the phase shift between estimated and actual back-EMF. The improved method demonstrates accurate speed and position estimation without performance loss, both in open-loop operation and under closed-loop speed control.

The proposed Improved Sensorless Approach has been validated under various operating conditions, including different speed ranges and load levels. The results confirm that the motor performance achieved with the improved sensorless strategy is comparable to that obtained using mechanical Hall-effect sensors. In addition, the method maintains low computational complexity, making it suitable for real-time implementation in cost-sensitive, high-precision industrial applications that require reliable sensorless operation.

While the proposed sensorless approach offers significant advantages in terms of dynamic tracking, its performance is inherently linked to the accuracy of the motor parameters, specifically the stator resistance and inductance. This research has quantified the robustness limits of the observer, demonstrating that the method maintains acceptable performance—with a position RMSE of approximately 8° and a speed MAE below 5%—for parameter variations up to 120% of the nominal resistance and 90% of the nominal inductance. However, beyond these thresholds, particularly under extreme thermal or magnetic saturation, the reconstruction of the back-EMF suffers from structural biases. It was observed that even with a fast observer dynamic (set to twenty times the electrical frequency) and a non-identical pole placement strategy, the estimation accuracy remains sensitive to model mismatches. These findings establish a clear operational envelope for

the current method and highlight the necessity for future research to integrate online parameter identification or adaptive estimation techniques to ensure reliability across all practical operating conditions.

Finally, experimental validation on a dedicated BLDC drive system will be conducted to confirm the effectiveness of the proposed approach under real-world constraints. Hardware implementation will facilitate a rigorous evaluation of the observer's resilience against measurement noise, inverter non-idealities, and switching effects. Most importantly, future experimental campaigns will focus on validating the identified robustness boundaries by subjecting the motor to various thermal and load profiles. This will provide a more comprehensive assessment of the applicability of the Improved Sensorless Approach in demanding industrial environments, ensuring that the estimated operational envelope (limiting errors within acceptable margins) is maintained in the presence of unmodeled hardware dynamics.

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